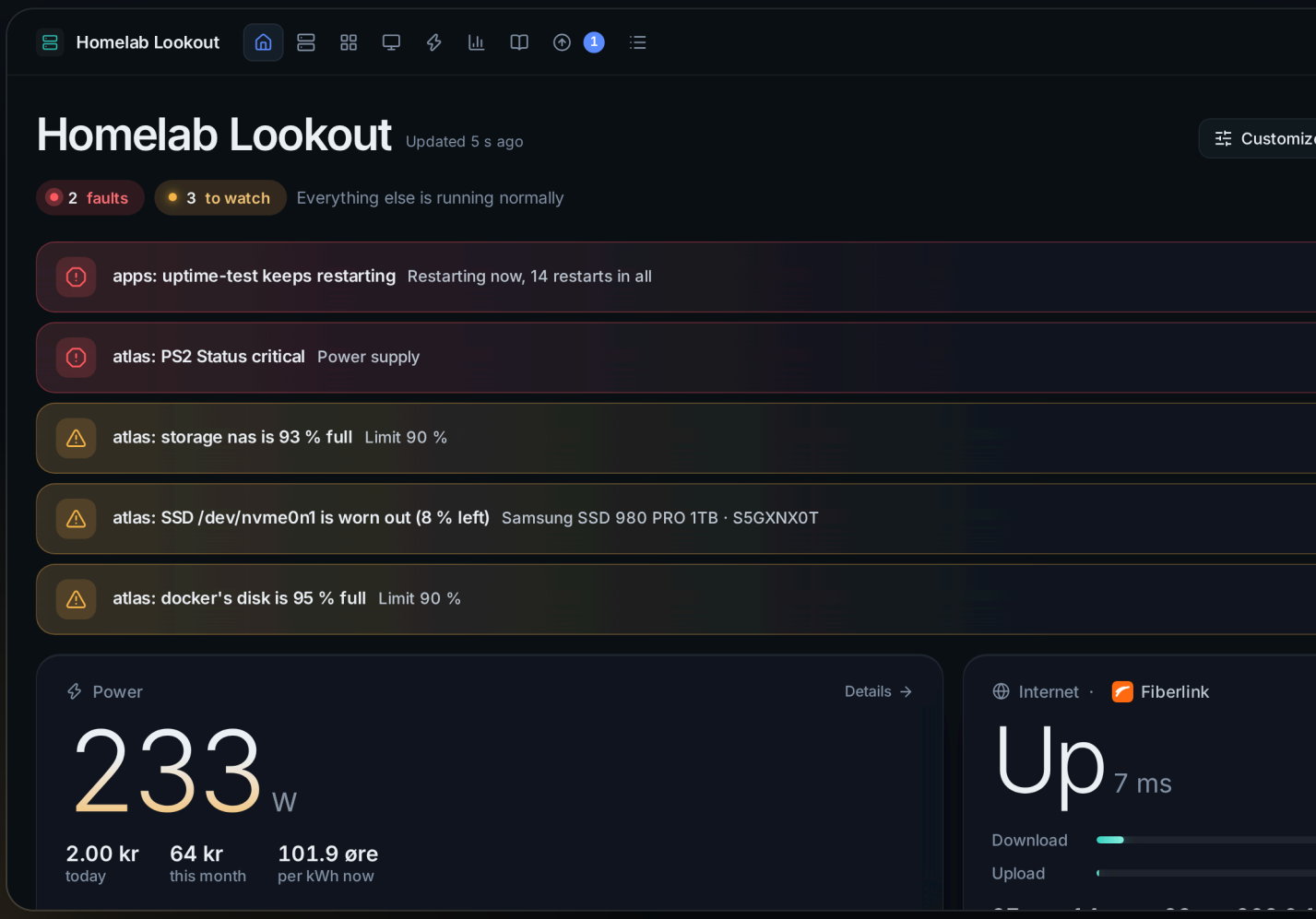
USER GUIDE

Homelab Lookout

Your servers, network, power and services on one calm screen. This guide walks through every page with a sample lab, from signing in to patching cables in the rack.



Version 2026.10.14 · All names, numbers and addresses in this guide are sample data.

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What's inside

Homelab Lookout reads your equipment and shows how it is doing. It never changes anything on a device: there is no button that restarts, switches off or updates hardware. You can leave it on a wall screen or hand the link to a family member without worrying.

Power
What the lab draws now, what it costs today and this month, with your own tariff.

Network
Internet up or down, traffic, clients, your provider and public IP.

Hardware
Every device drawn as it looks, with its live lights, problems and history.

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12 **Admin: alerts**
Push, ntfy and more; acknowledging

13 **Admin: check-ins and maintenance**
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14 **Admin: dashboard, look, power**
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15 **Admin: backups, security, server**
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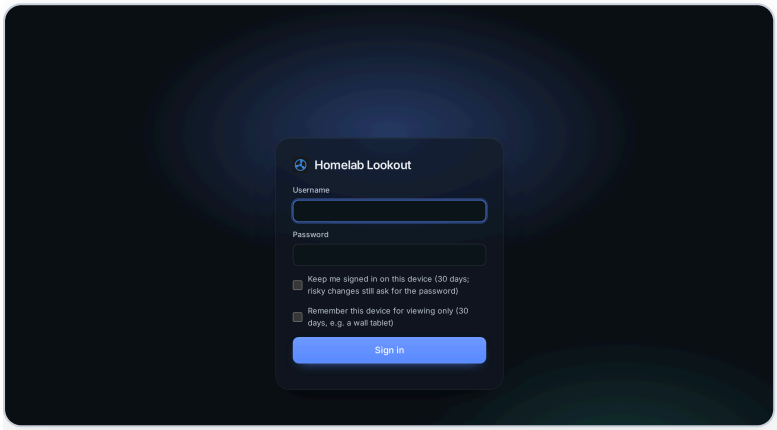
17 **When something doesn't work**
Common problems and fixes

About the sample lab. The screenshots show an example setup: a UniFi gateway (*edge*), a 24-port switch (*core-switch*), two Dell servers (*atlas* with Proxmox, *vault* with TrueNAS), a Docker host (*apps*), a patch panel, an access point, two cameras and twelve services. A power supply in *atlas* is failing and a container keeps restarting on purpose, so you can see how problems look.

CHAPTER 1

Getting started

Lookout runs on one machine in your lab and you open it in a browser, for example `https://lookout.home.arpa:8443`. Ask whoever installed it for the address and the admin login.



Sign in. *Keep me signed in* on your own phone and computer; *Remember this device for viewing* on a wall tablet.

Sign in

Enter the admin user name and password. After five wrong passwords you have to wait, first 30 seconds, then longer each time, up to an hour.

- A normal sign-in lasts 12 hours (2 hours without use).
- **Keep me signed in**: 30 days, ending after a week without use. Restarts and updates don't sign you out.
- A few risky changes ask for the password again if you haven't typed it in the last 15 minutes: backups, alert channels, the web server and making things public.

Passkeys

Once you have added one (chapter 15), **Sign in with a passkey** uses Face ID, Touch ID, Windows Hello or a security key. The password keeps working. Passkeys work on the dashboard's domain name, not on its IP address.

Remember this device for viewing

For 30 days, that device can **look** at every page without the password, but it can't change anything. Use it for a wall tablet.

Trust the certificate (once per device)

Lookout makes its own small certificate authority, so the browser warns you until the device trusts it. Download it from `/ca.crt` or **Admin → Server → Download CA certificate**, check the fingerprint against the one shown under **Certificate**, and import it:

Device	How
iPhone / iPad	Open the file → Settings → *Profile Downloaded* → Install. Then General → About → **Certificate Trust Settings** → turn it on.
Android	Settings → Security → Encryption & credentials → Install a certificate → **CA certificate**.
Windows	Double-click → Install Certificate → Local Machine → *Trusted Root Certification Authorities*.
macOS	Double-click → Keychain Access (System) → open the certificate → Trust → *Always Trust*.
Linux	Copy to `/usr/local/share/ca-certificates/` and run `sudo update-ca-certificates`.

The certificate authority can only vouch for this one server, so trusting it doesn't let it pretend to be other websites.

Install it as an app

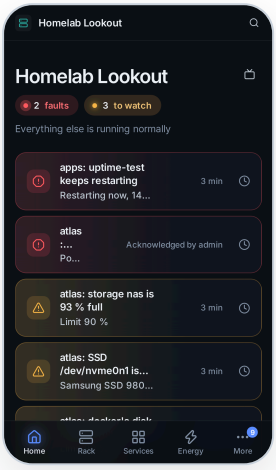
1. Trust the certificate first (above).
2. **Android:** Chrome menu ⋮ → **Install app**. **iPhone:** Safari → Share → **Add to Home Screen**. **PC:** the install icon at the end of the address bar.
3. Open the app and sign in with *Keep me signed in* ticked. On an iPhone, sign in *inside* the app.
4. For alerts on the phone, turn on **push** in the app (chapter 12).

The app icon is your logo (chapter 14). When the server can't be reached, the app says *Can't reach the dashboard* and opens by itself once it is back.

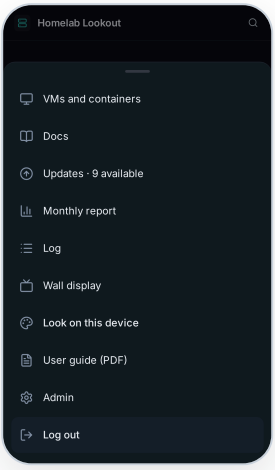
CHAPTER 2

Finding your way

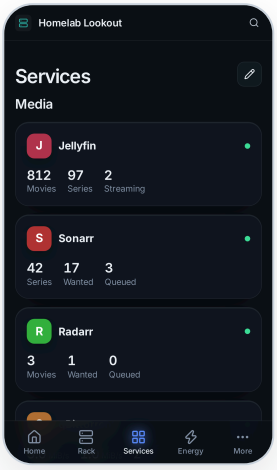
The pages are in the bar at the top: **Home · Rack · Services · VMs · Energy · Docs · Updates · Log**. On the right are Search and your avatar: click it for **Admin**, **Look on this device**, the **User guide** and **Log out**.



Phone. The tab bar at the bottom has Home, Rack, Services and Energy.



More has VMs, Docs, Updates, Log, the wall display, the guide, the look, Admin and Log out.

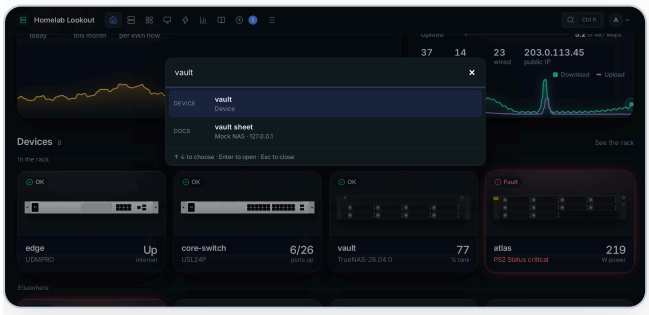


Every page works on a phone, one column at a time.

Docs and Updates appear when you're signed in or on a remembered device. Log and Admin are for the admin only.

Search

Press `Ctrl K` or `/` anywhere. It finds devices, services, VMs, Docs sheets and pages (including the text inside them), IP reservations and settings. Use ↑ ↓ to choose, **Enter** to open and **Esc** to close.



Search for *vault*: the device and its Docs sheet.

Your own look

The **palette** button changes the look **on this device only**: Auto, Dark, Light, more themes and an accent colour. Everyone else keeps the owner's look from Admin. **Use the owner's look** goes back.

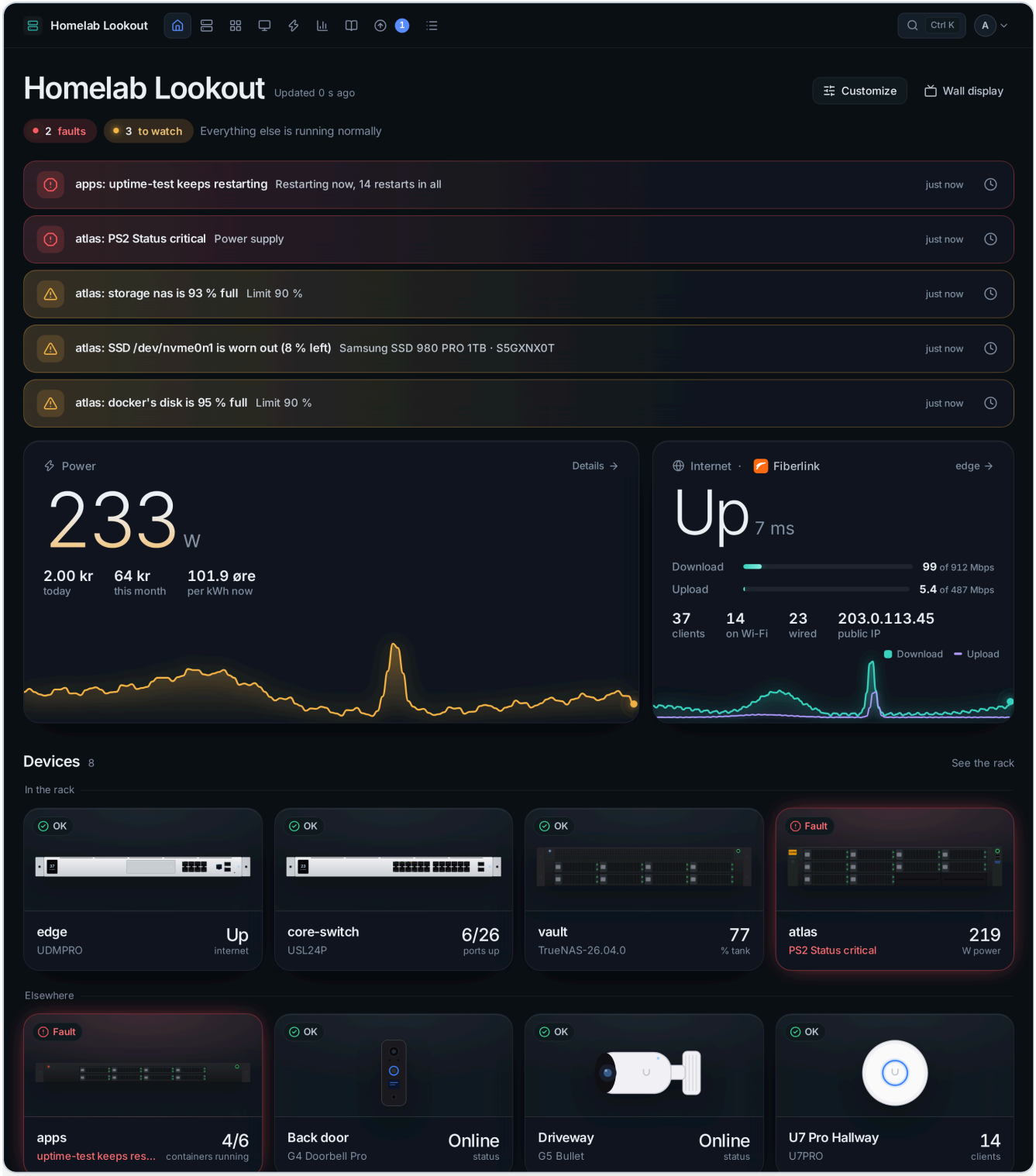


The look menu, for this device only.

CHAPTER 3

Home

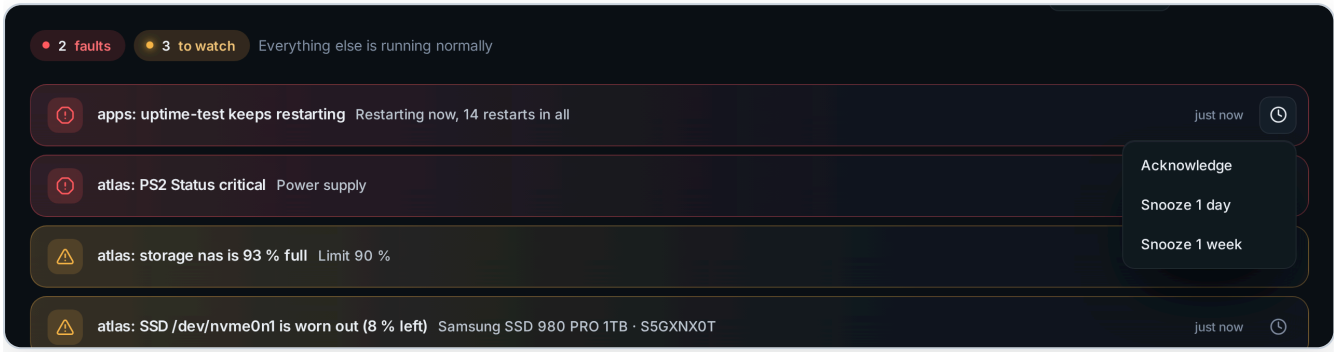
Home answers one question: is everything all right? It shows problems first, then power and the internet, then every device drawn the way it looks.



Home in the sample lab: the problems, the power and internet cards, and the devices.

At the top: the summary and "Needs you"

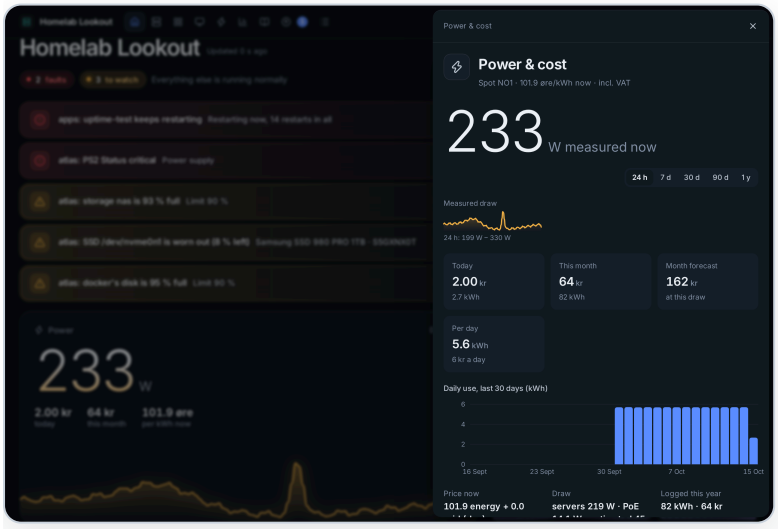
The pill says how many faults and things to watch there are, or **All normal**. Below it, one row per problem, worst first, with how long it has been going on. Click a row to open the device. A connection that stops answering is listed only after 2 minutes, so a single slow reading doesn't raise an alarm.



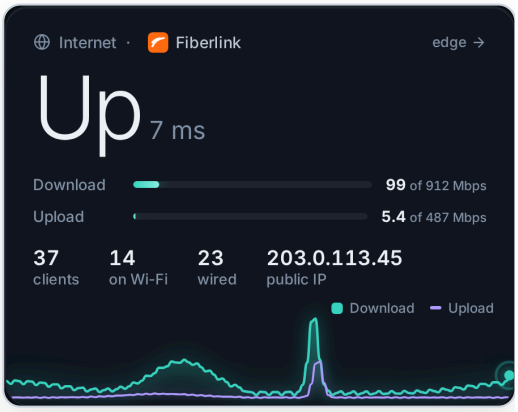
The clock button on a row: *Acknowledge* it, or snooze it for 1 day or 1 week.

- **Acknowledge** (admin): someone is on it. The problem stays until it clears, but no more reminders are sent. The row says *Acknowledged by …*. You can also acknowledge from the alert message itself (chapter 12).
- **Snooze** (admin): the problem leaves the top of Home and the wall display for 1 day or 1 week, but the device still shows it and alerts are still sent. A warning that turns into a fault comes back by itself.
- **Behind core-switch**: when a switch or host is down, what is reached through it is grouped under it, muted, with one alert for the cause (chapter 10).
- **Maintenance** and **Alerts muted**: problems during planned work, or on something whose alerts are switched off. Shown, not sent.
- **Public IP changed**: shown until an admin closes it. It stays in the Log.

Power and Internet



Power details: today, this month, the forecast, cost per day and the yearly figures.



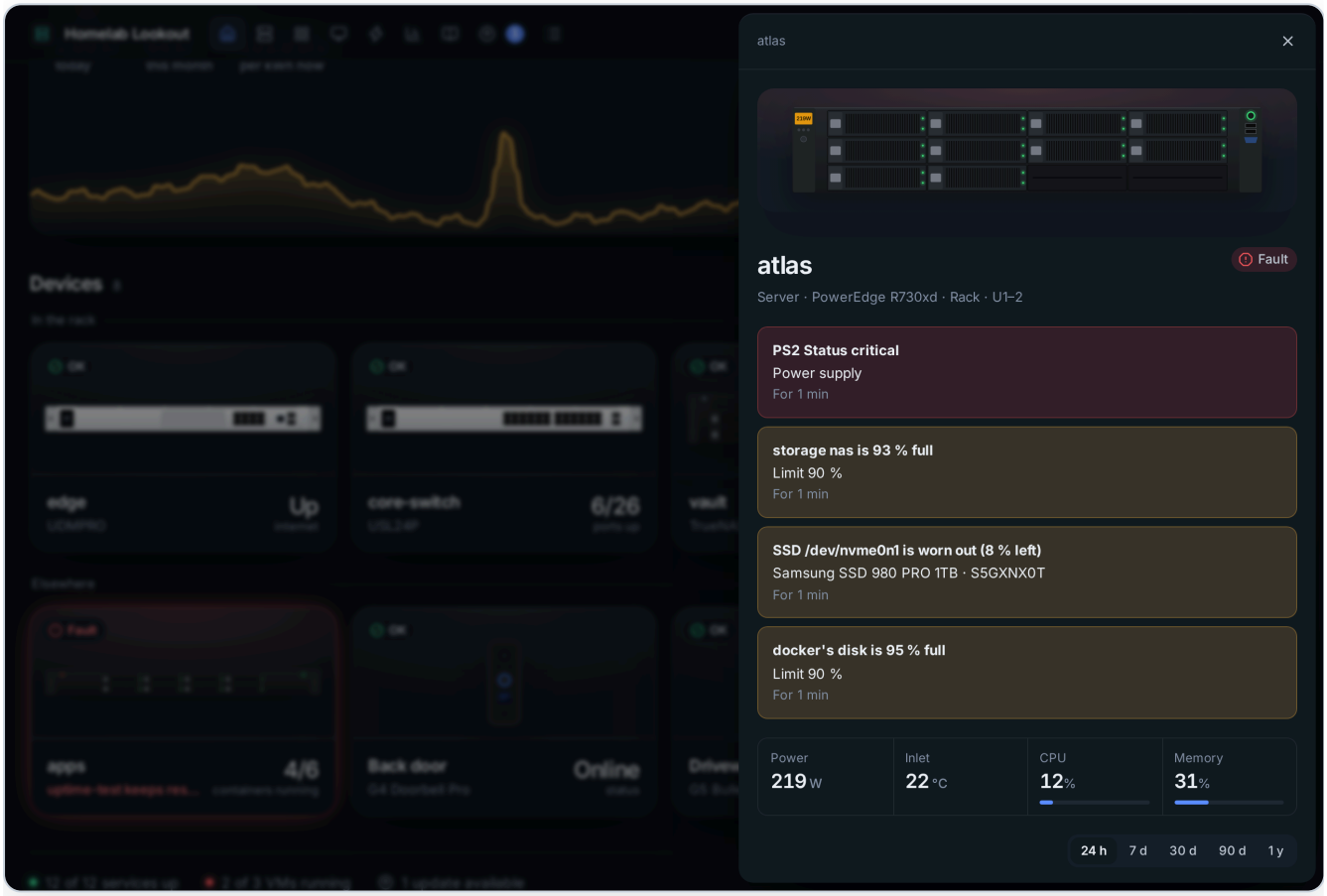
Internet: traffic now against your speed test, the last 24 hours, clients, provider and public IP.

Power shows what the lab draws now, today's and this month's cost, and a graph of the last 24 hours. Power comes from server controllers, smart plugs, UPSes and PoE, and estimates for the rest. Click it for the details and the Energy report.

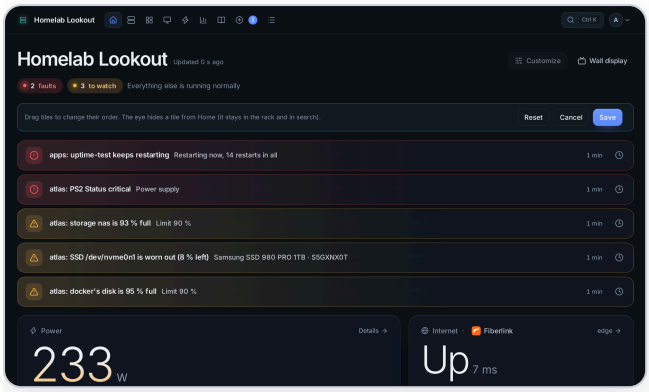
Internet comes from your gateway (UniFi or Omada), with your provider and its logo. The public IP is only shown to people who are signed in.

Devices and their details

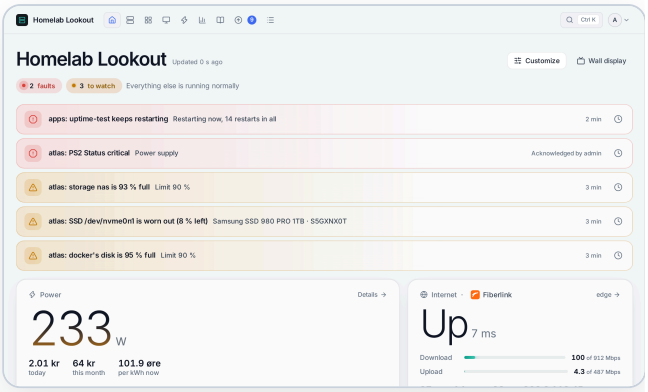
Each device has a tile with its drawing, one status (**OK**, **Warning**, **Fault**, **Offline**, **Unreachable** or **Maintenance**), its worst problem or model, and one key number. Click a tile for the details: problems first, then numbers, graphs, power supplies, fans, drives, ports, VMs and containers, with buttons for its Docs sheet and Admin.



Details for atlas: the failing power supply first, then power, temperature, CPU and memory with their graphs.



Customize (admin): drag tiles into another order and hide the ones you don't need.



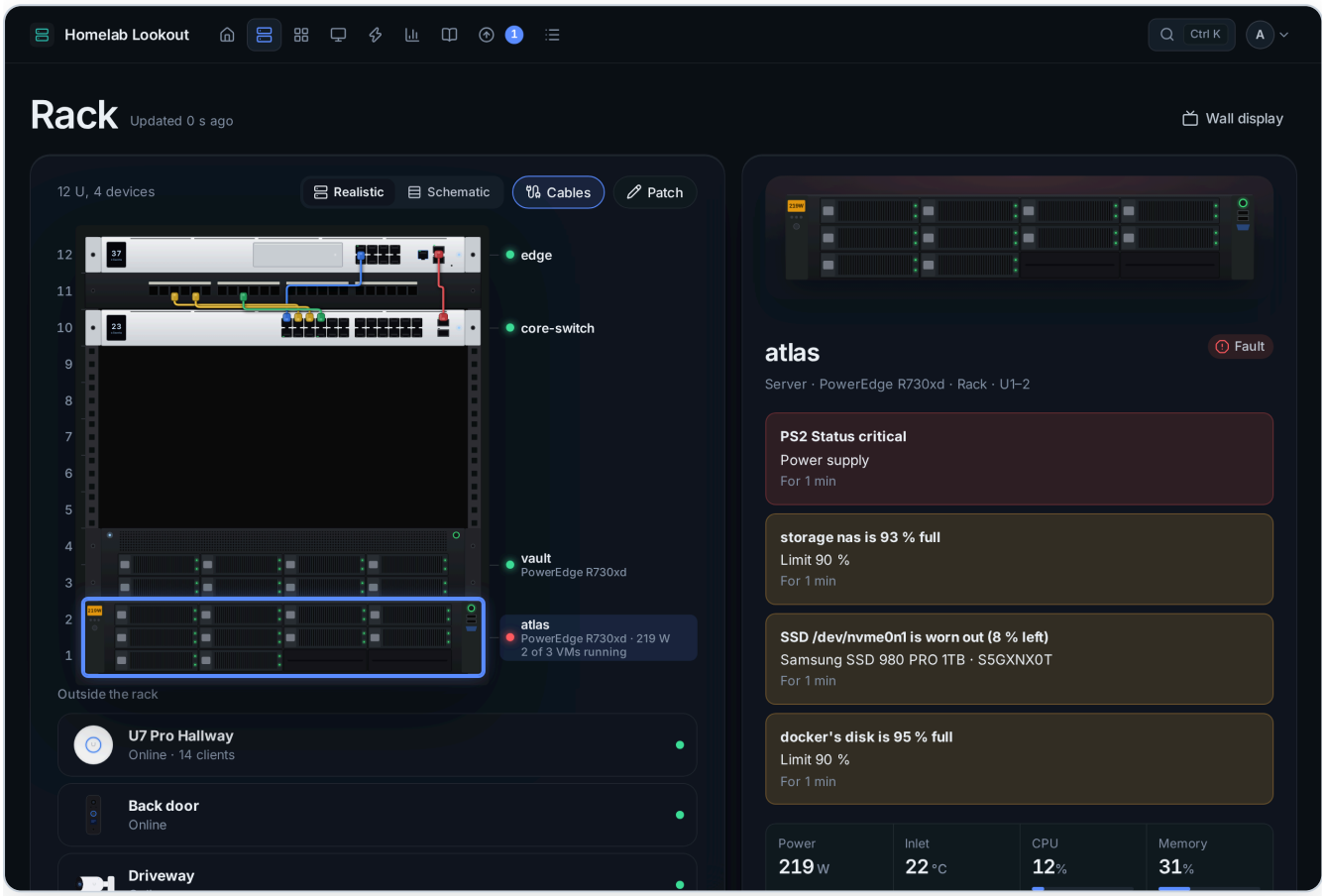
The same Home in the light theme. Every theme has the same layout.

One quiet line at the bottom counts services up, VMs running and updates available. Each is a link.

CHAPTER 4

Rack and patch cables

The rack is drawn to scale with live lights: the drives your servers really have, switch ports lit by link and PoE, status LEDs. A device with a problem gets a red or amber outline.



The Rack page with cables shown. On a wide screen, the chosen device's details stand beside the rack.

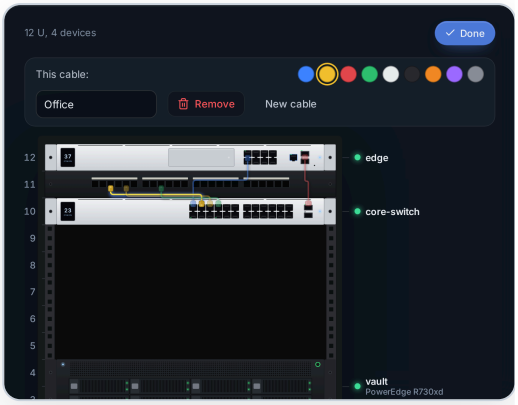
- **Realistic / Schematic** switches between the drawing and plain blocks with names, status and power.
- Click a device to see its details. Devices outside the rack (access points, cameras) are listed below it.

Patch cables

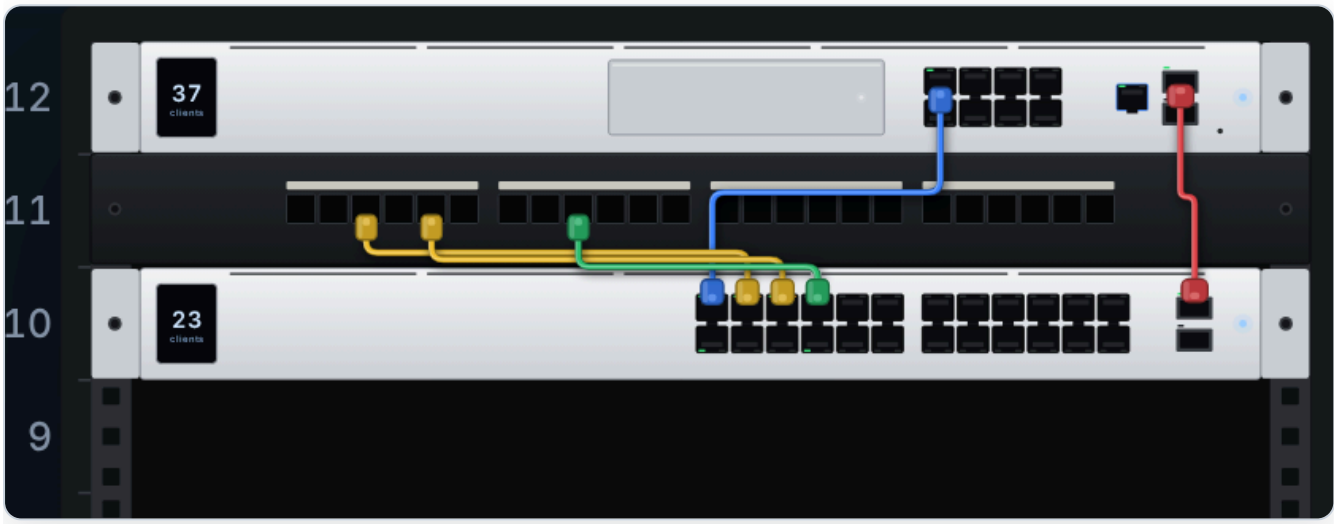
Cables shows or hides the patch cables in your browser. They are off at first, so the rack stays clean.

To draw them, sign in and press **Patch**:

1. Pick a colour in the bar above the rack.
2. Click a port on a switch, gateway or patch panel.
3. Click the port at the other end. The cable is saved.
4. Click a cable to change its colour, add a label, or **Remove** it.
5. **Done** leaves patch mode.



Patch mode with a cable picked: its colour, label and Remove.



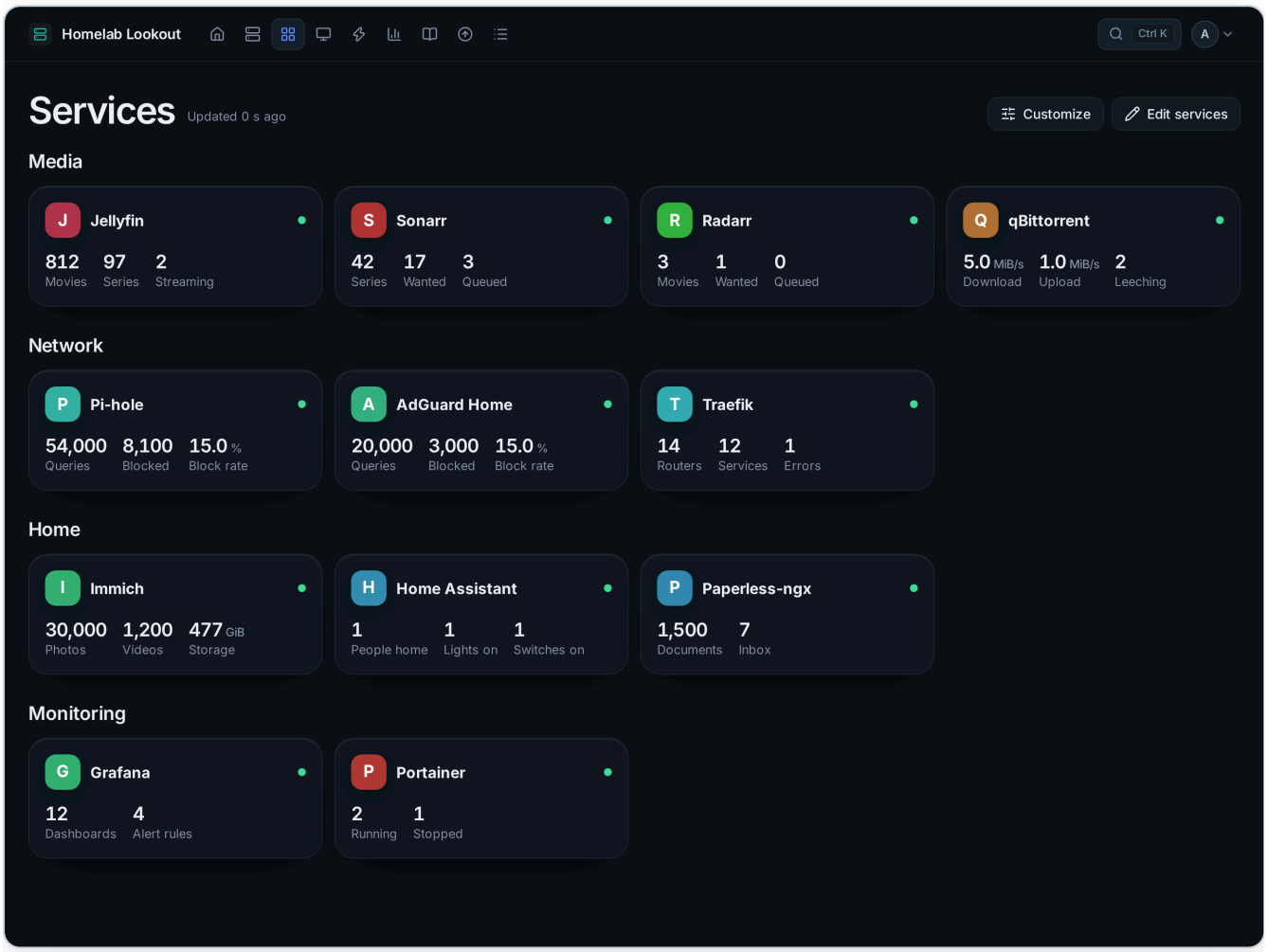
Close-up: the gateway to the switch (blue, red for the 10G link) and three runs from the switch to the patch panel.

> Every cable change is written to the Log. Removing a device or patch panel from the rack also removes its cables. The wall display shows cables when you tick *Patch cables* in its Customize panel.

CHAPTER 5

Services and the status page

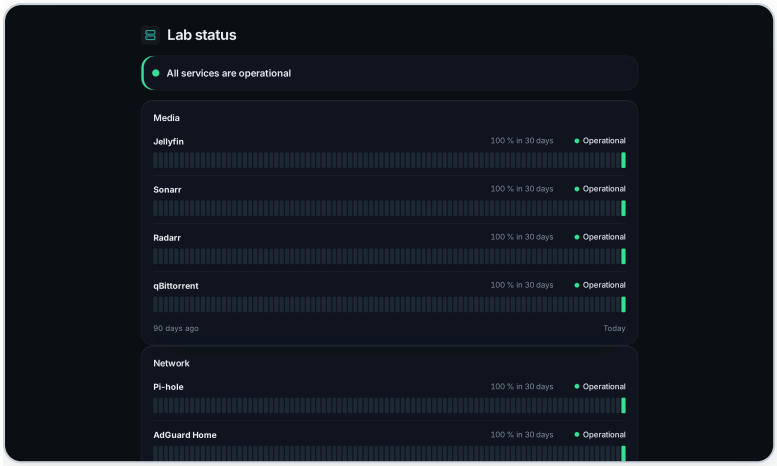
Your self-hosted apps, grouped the way you like, each with a status dot and up to four live numbers. Click a card to open the app.



Services in four groups, each with its status and numbers.

- The dot is green when the service's **status check** passes: a web page, a ping, a TCP port or a DNS lookup (chapter 11). Point at it for the response time.
- For an `https://` service the certificate's expiry is watched too: a warning two weeks before, critical the last three days.
- The numbers come from each app's integration: movies and series in Jellyfin, queued downloads, blocked queries, photos in Immich, and so on.
- **Customize** (admin) drags services between groups, renames groups, sets columns and hides services.

The status page



The status page at `/status`: up or down and the uptime of each service, with no links or addresses.

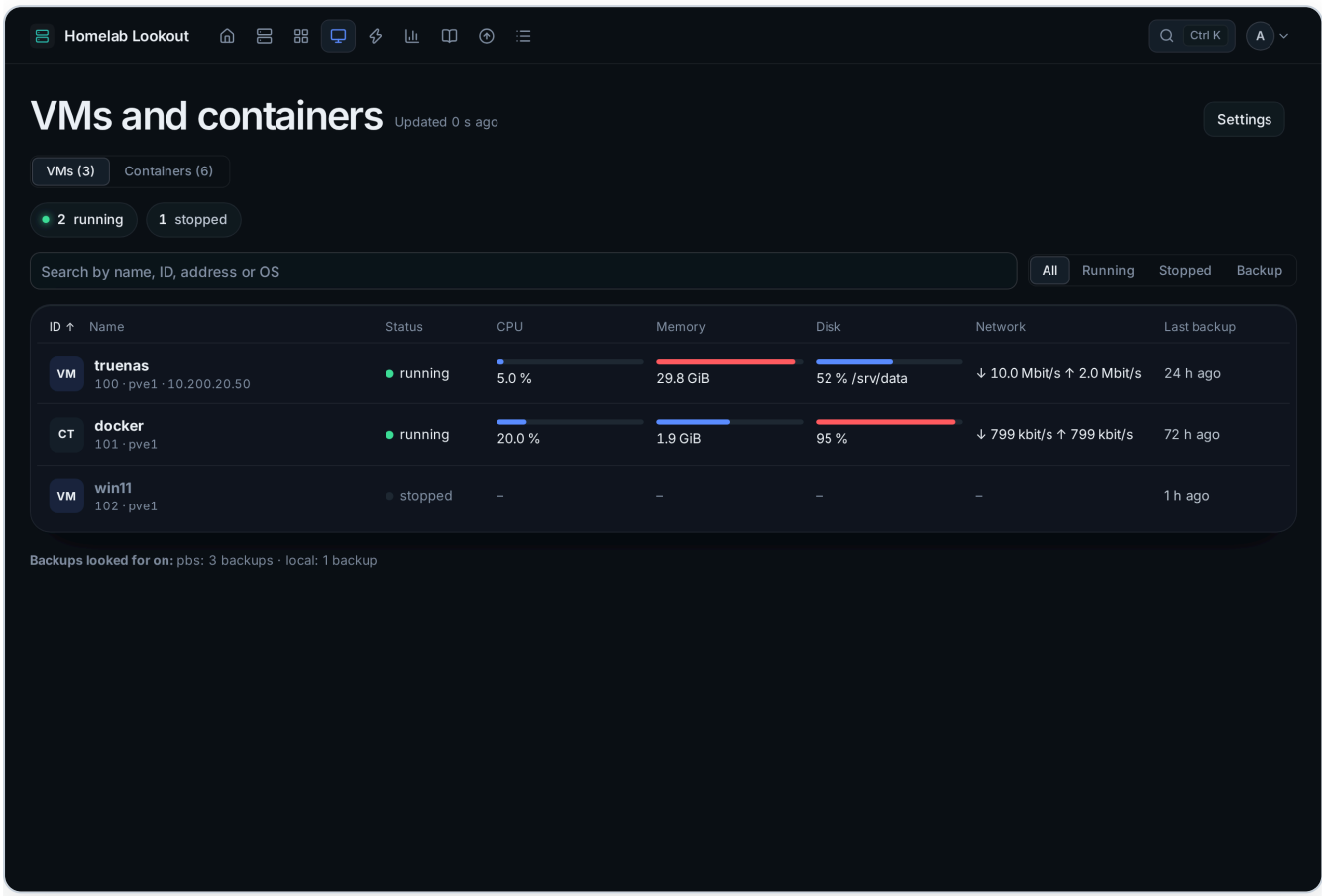
A public page for the people who use your services: whether each one is up, its uptime over 90 days (30 on a phone), and planned maintenance. It shows no links, addresses, devices or VMs.

Switch it on in **Admin → Dashboard**. It works while the dashboard itself stays private, so you can share it instead of the dashboard.

CHAPTER 6

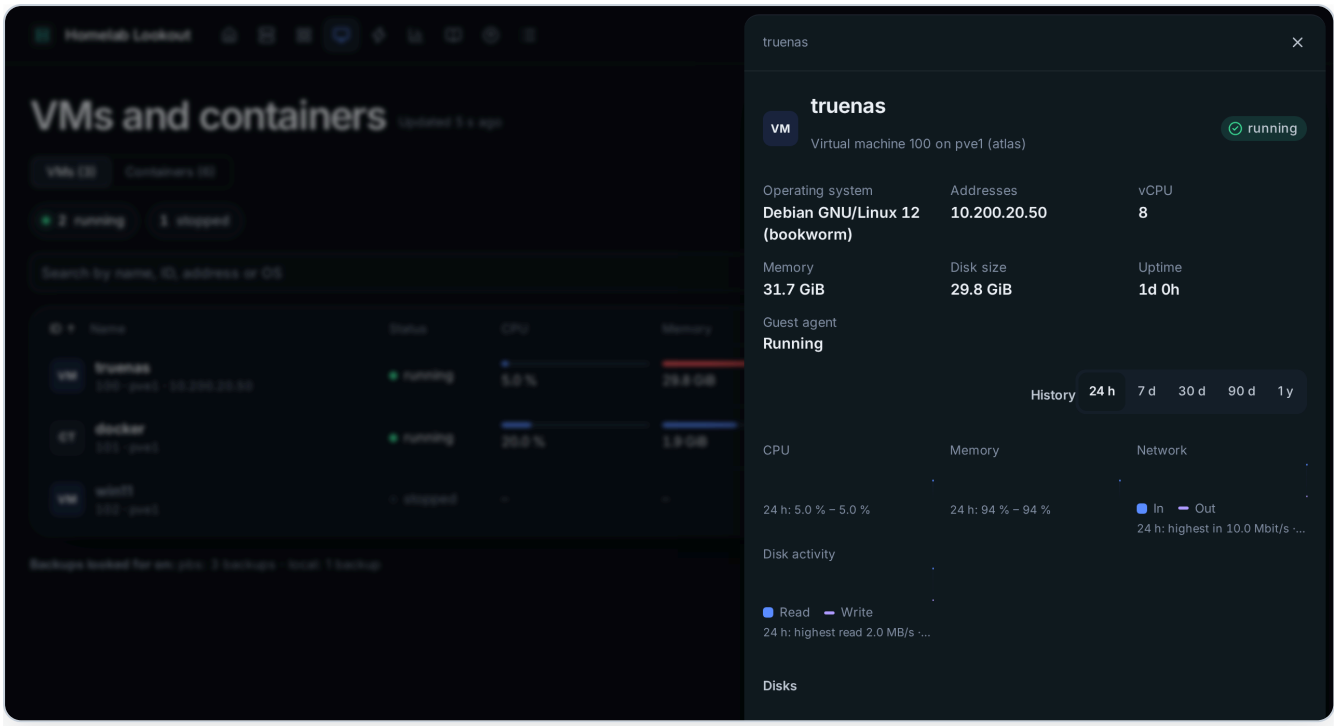
VMs and containers

Every VM and container on your Proxmox hosts, and every Docker container, on one page: **VMs** in the top bar.



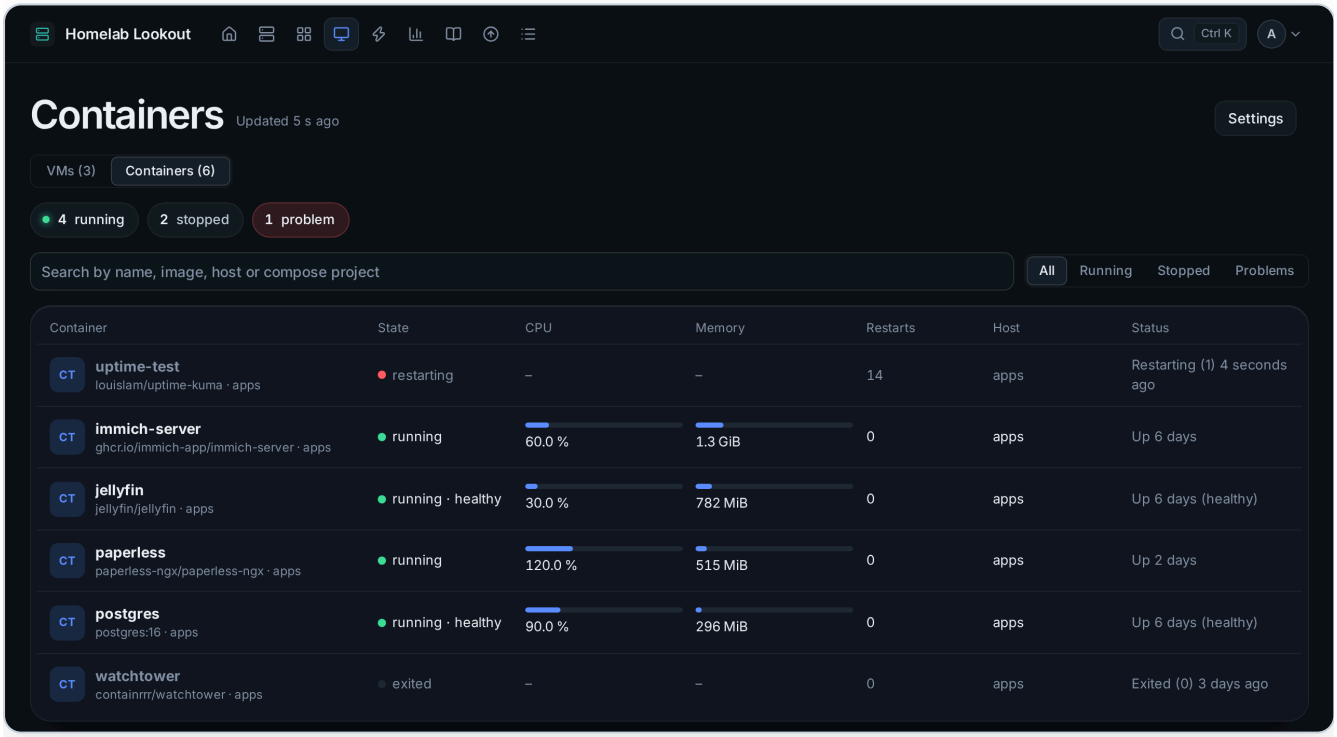
VMs: status, CPU, memory, disk, network and last backup for each guest.

- **Search** by name, ID, address or OS, and filter **Running**, **Stopped** or **Backup** (no backup, or an old one). Click a column heading to **sort** (by ID at first). **Settings** opens where VMs are set up.
- The disk is a container's own disk, or for a VM with the QEMU guest agent its fullest filesystem. The address comes from the guest agent too.
- **Last backup** is the newest backup Proxmox has of the guest, on any backup storage. A guest is **overdue** when its newest backup is older than its backup job's interval plus a day (a monthly job allows 32 days); that raises a *Backup failed* warning.



A guest's details: size, graphs of CPU, memory, network and disk, its filesystems, its backups and backup job, and *Open in Proxmox*.

Containers



Containers from a Docker host: state, health, CPU (100 % is one core), memory, restarts and Docker's own status.

Containers come from a Docker host through a read-only socket proxy, or from Portainer (chapter 10). A Portainer card under Services has *Show its containers*, which adds that connection for you. A container that **keeps restarting**, one that **stopped** although it should run, or one that fails its **health check** shows under *Needs you* and sends a *Container problem* alert. A container you stopped by hand doesn't count, and the label `lookout.ignore=true` leaves one out.

CHAPTER 7

Energy report

How much the lab used and what it cost, per day or per month, split per device, priced with your own electricity plan and grid tariff.



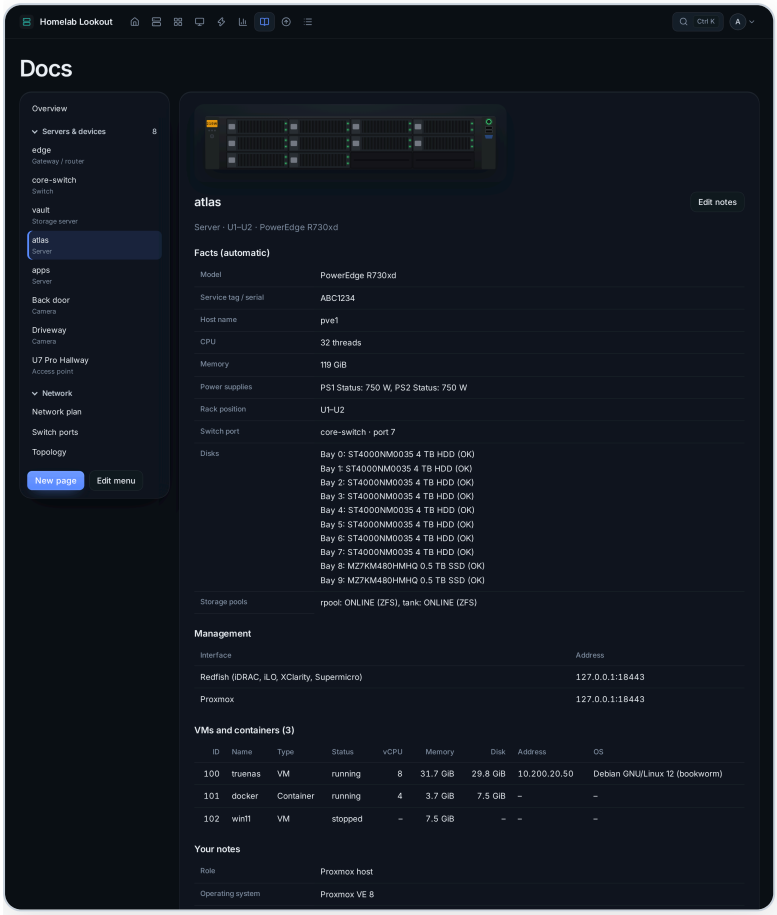
October in the sample lab: kWh, cost and average price, a bar per day, and each device's share.

- **Month / Year** switches between days in a month and months in a year; ‹ › step through periods.
- Point at a bar for that day. **Show as table** gives the same numbers as a table.
- **Per device** separates measured power (server controllers, smart plugs, PoE) from estimates you entered.
- **Downloads:** *Excel (NO)* opens straight in Norwegian Excel; *CSV* is for everything else.

CHAPTER 8

Docs, Updates, Log and the report

Docs

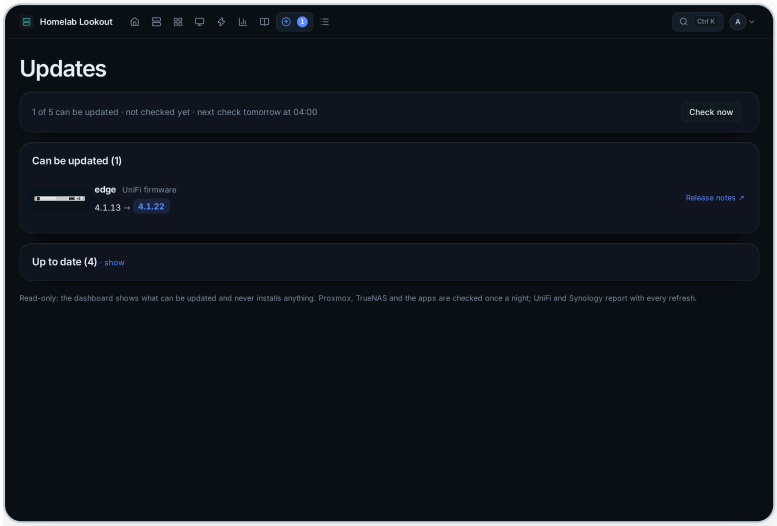


The server sheet for atlas: facts read automatically, then your own notes.

- **Server sheets:** model, serial, CPU and memory, disks, rack position, addresses and switch port fill in by themselves. **Edit notes** adds your own.
- **Network plan:** networks and IP reservations, importable from UniFi.
- **Pages:** your own pages in Markdown, with a history of the last 50 versions and *Recently deleted*.
- **Edit menu** arranges the menu in groups.

> Docs are not for passwords. Link to your password manager instead.

Updates

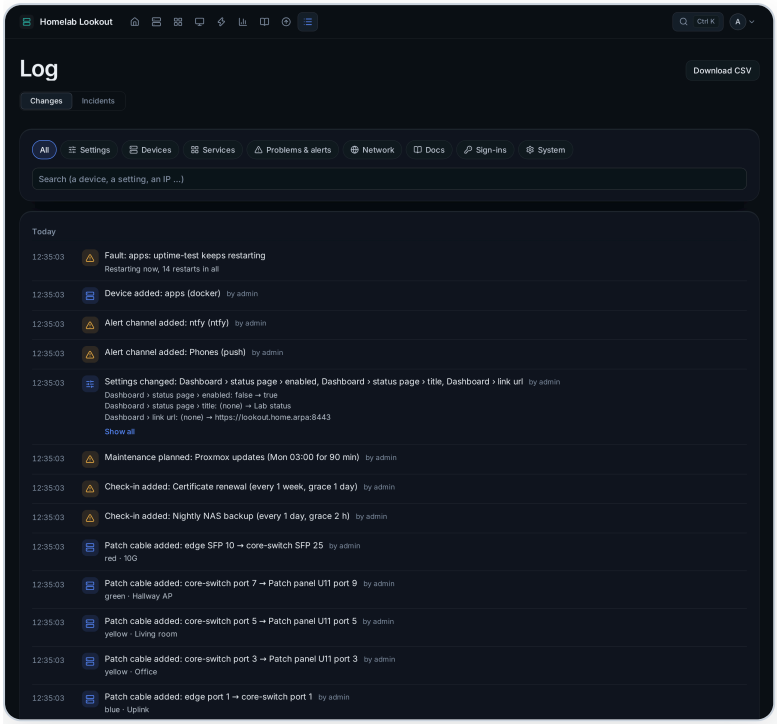


Updates: installed and new versions, with release notes.

One list for Proxmox, TrueNAS and its apps, UniFi firmware, Synology and most services. It is checked every night; **Check now** checks right away.

Nothing is ever installed from Lookout. Update on the device itself, then press *Check now*.

Log

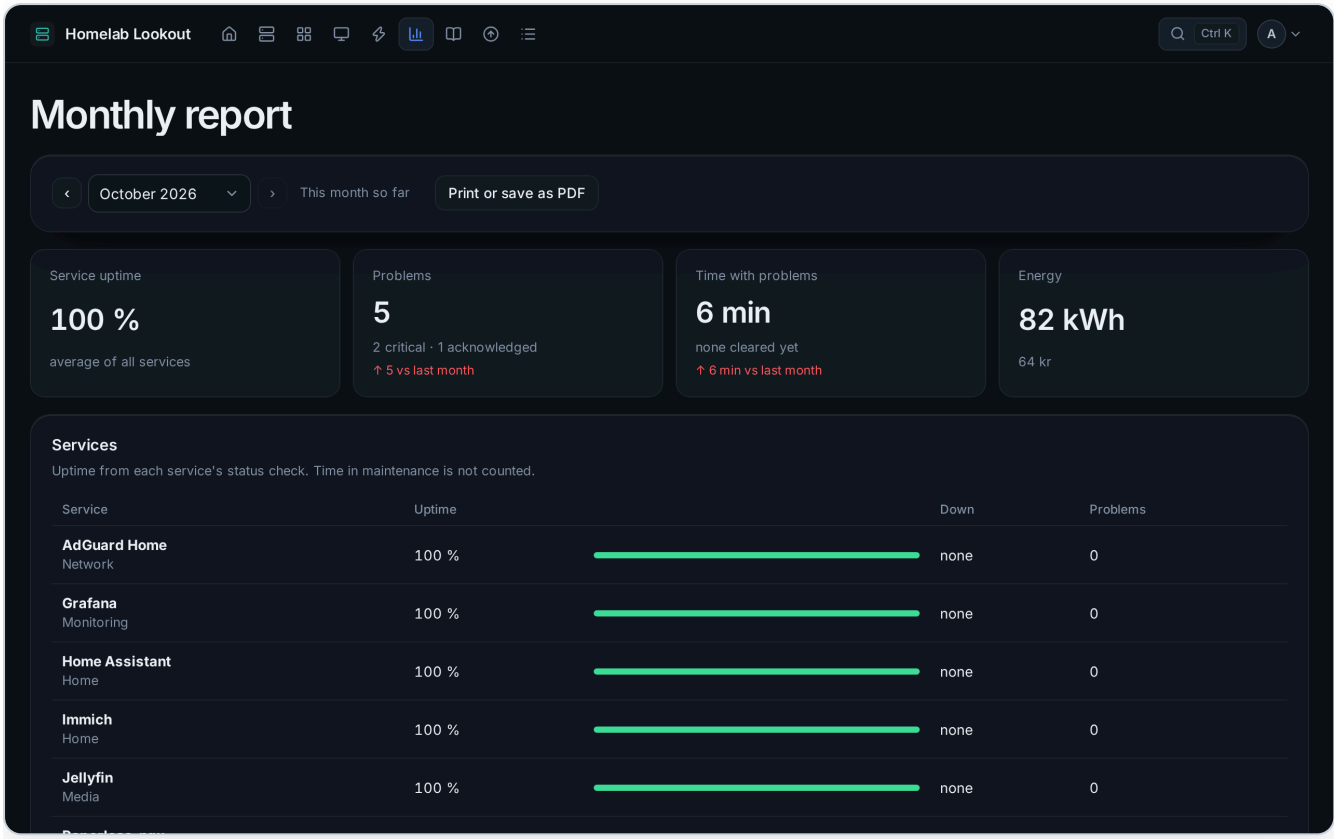


The Log: every change, with what changed and who did it.

Everything that happens, newest first, grouped by day: changes in Admin and Docs (*Title: Homelab → My Lab*), devices found, problems starting and clearing, notifications sent, acknowledgements, snoozes, IP changes, patch cables, sign-ins, update checks and backups.

Filter by category, search, and **Download CSV**. Entries are kept for a year and go into backups. Passwords are never written to it.

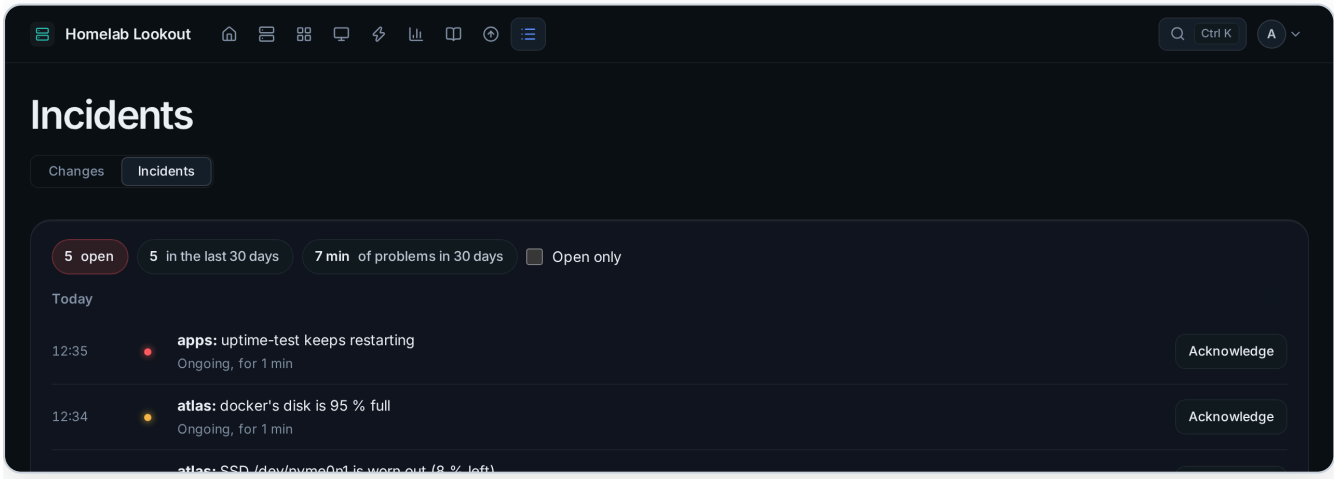
Monthly report



The monthly report: uptime, problems and energy against last month, then services, devices, the longest problems and storage.

Report in the top bar sums up a month: the uptime of every service and device, the problems and how long they took to clear, energy and cost, and how each storage grows with when it will be **full** at that pace. A storage that will be full within 30 days also shows under *Needs you*. **Print or save as PDF** makes a clean copy.

Incidents

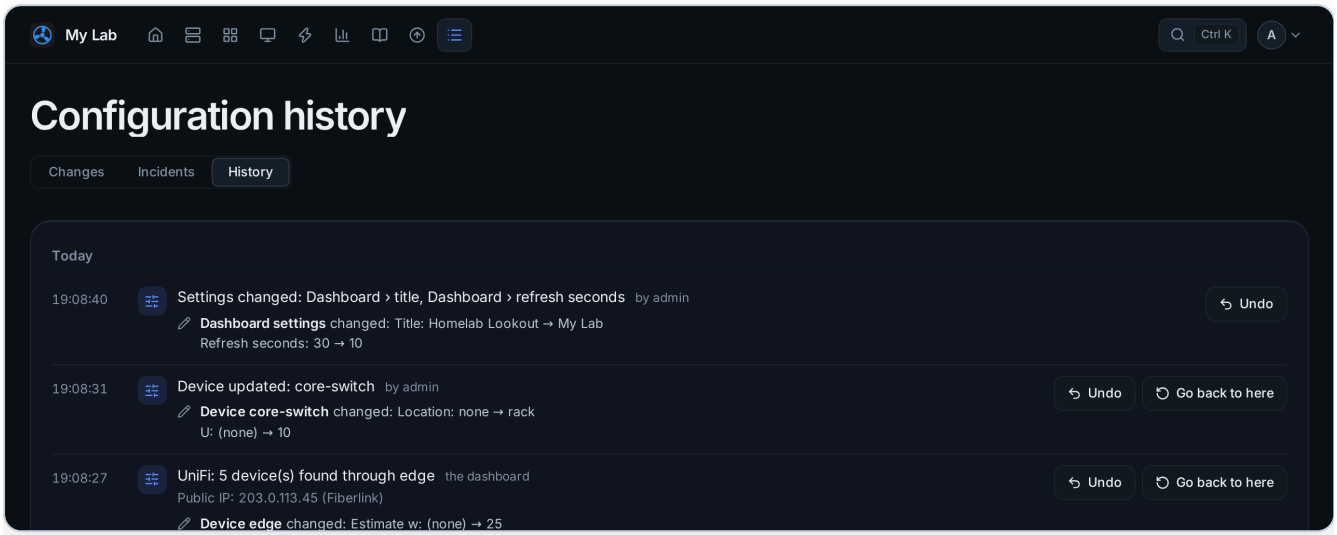


Incidents: each problem from start to end, who acknowledged it, and the totals for the last 30 days.

Incidents at the top of the Log lists every problem from when it appeared to when it cleared: how long it lasted, who acknowledged it and when, and whether it was behind another outage or during

maintenance. **Open only** shows the ones still going on, each with an **Acknowledge** button. Incidents are kept for a year.

Configuration history



History: every change to the configuration, with what it changed and buttons to undo it.

History at the top of the Log lists every change made in Admin, with who made it and what it changed. **Undo** puts back only what that change touched; a deleted device comes back with its connections and stored passwords. If something was changed again since, it says what and asks first. **Go back to here** puts the whole configuration back as it was right after that change. An undo can be undone too, and both ask for your password if you haven't typed it in the last 15 minutes. The admin password, passkeys and the web server's ports are never part of it, so an undo can't lock you out. Versions are kept for 90 days on the server.

CHAPTER 9

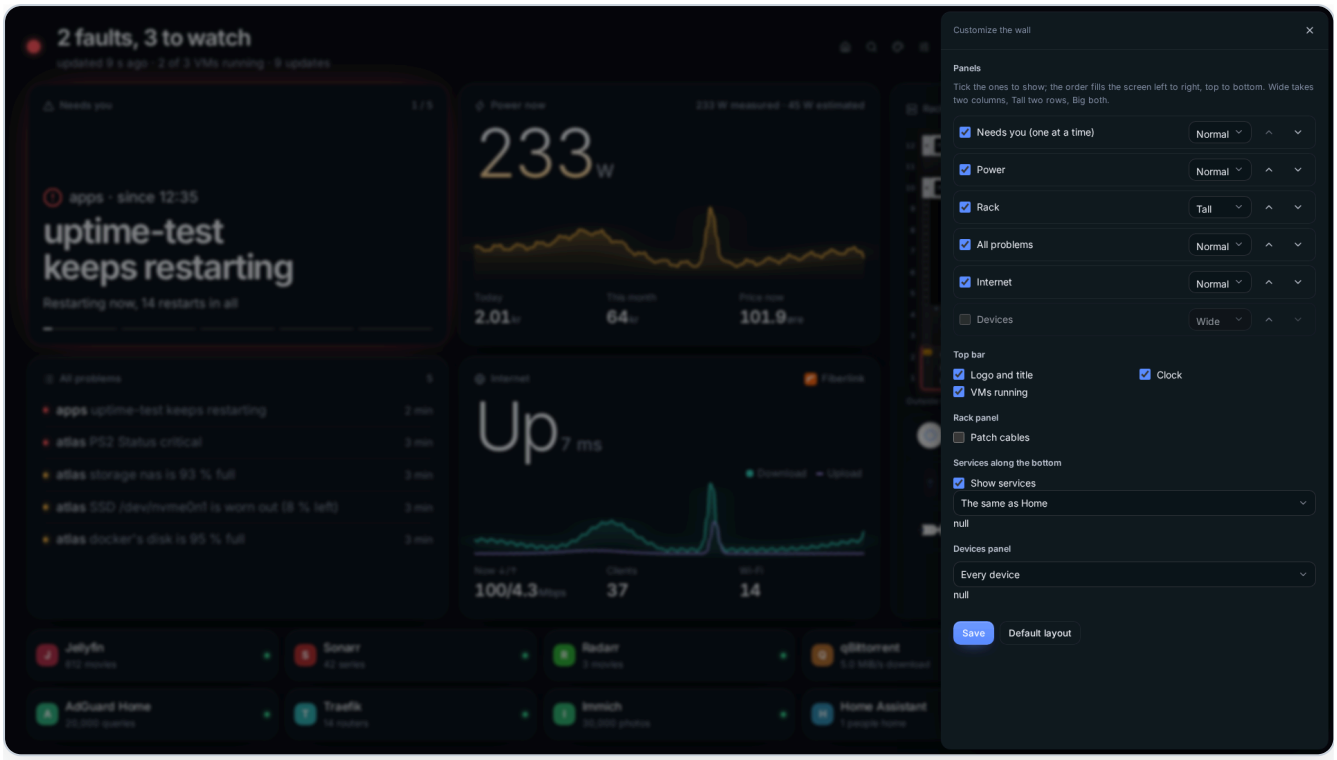
Wall display

One screen for a tablet or TV, readable from across the room. Open it from **Wall display** on Home, or at `/wall`.



The wall at 1920 × 1080: the headline and logo, the problem in focus, power, internet, the rack and the services along the bottom.

- **Needs you** shows one problem at a time, eight seconds each, with the full list beside it.
- The buttons by the clock fade out when nobody touches the screen; touch it or move the mouse to bring them back.
- Tap anything to open it on Home. The full-screen button keeps the screen awake where the browser allows it.
- For a wall tablet, sign in once with *Remember this device for viewing*; it then shows the wall without a password and can't change anything.

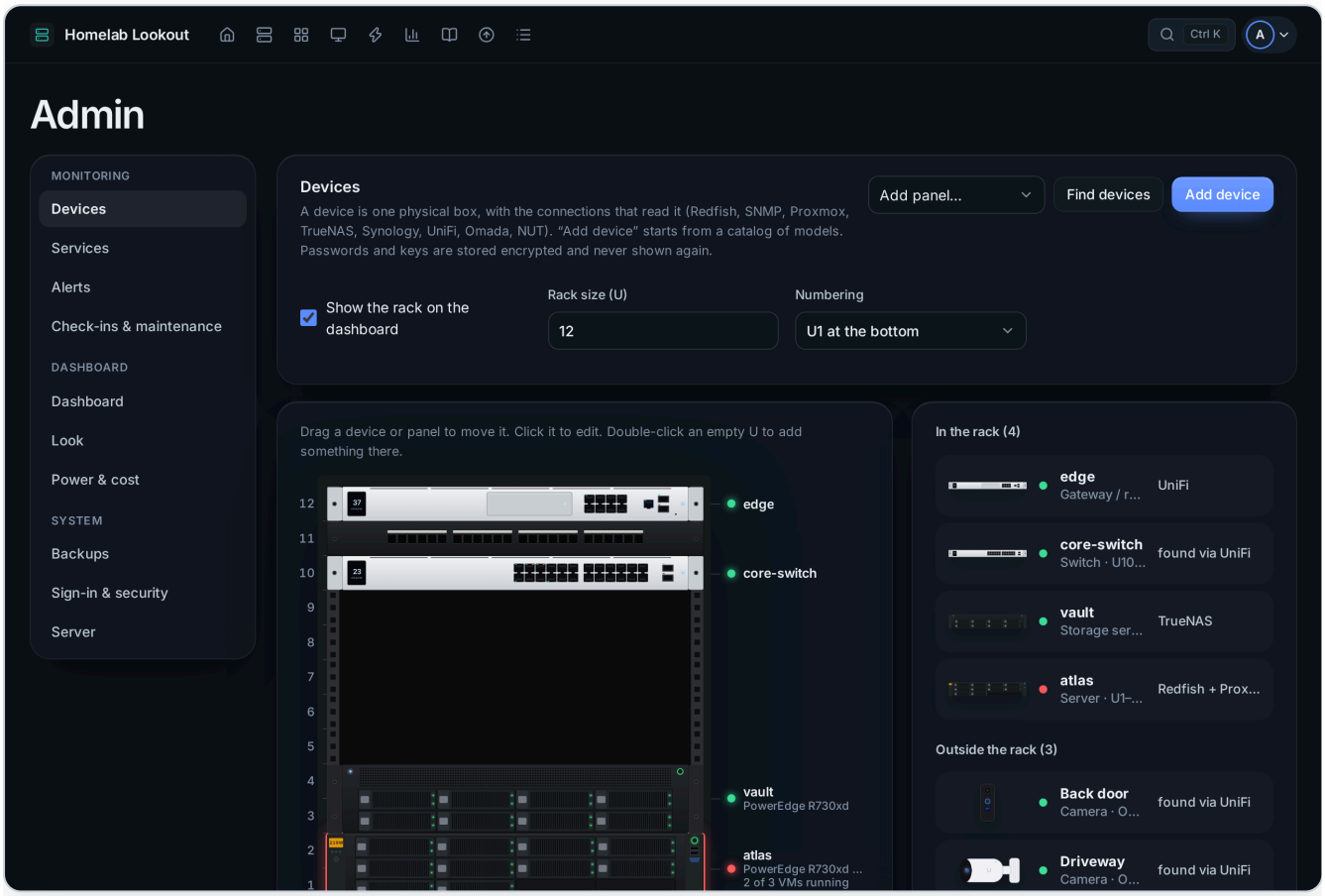


Customize (admin): choose panels, their order and size (Normal, Wide, Tall, Big), the top bar, patch cables, and which services and devices to show.

CHAPTER 10

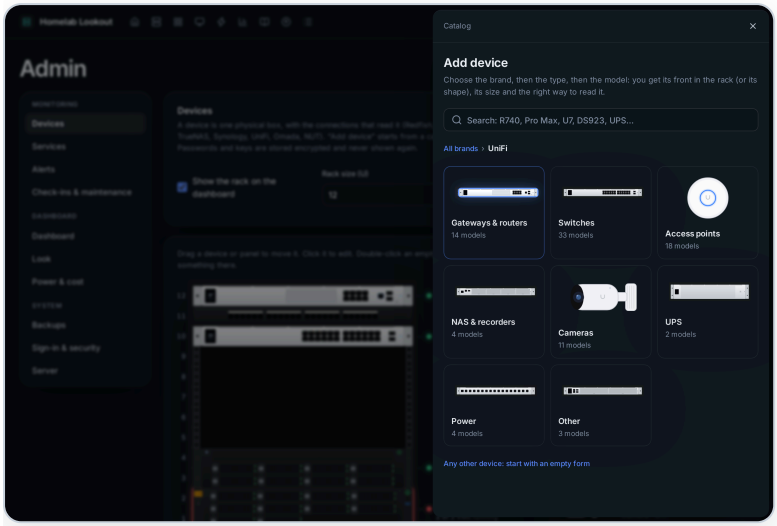
Admin: devices

Admin has ten tabs in three groups: **Monitoring** (**Devices · Services · Alerts · Check-ins & maintenance**), **Dashboard** (**Dashboard · Look · Power & cost**) and **System** (**Backups · Sign-in & security · Server**). On a phone, the section menu at the top switches between them. Passwords, tokens and keys are entered here and nowhere else; they are stored encrypted and never shown again.



Devices: the rack editor (drag to move, double-click an empty U to add a panel) and every device.

Add a device from the catalog



The catalog: brand, type, model. About 165 models, each drawn as it looks.

1. Press **Add device**.
2. Choose the **brand**, then the **type**, then the **model**. Search works from any step.
3. **Add this device** fills in the size, front, drive bays, typical power and the right connection.
4. Give it a name, fill in the connection, press **Test**, then **Save device**.

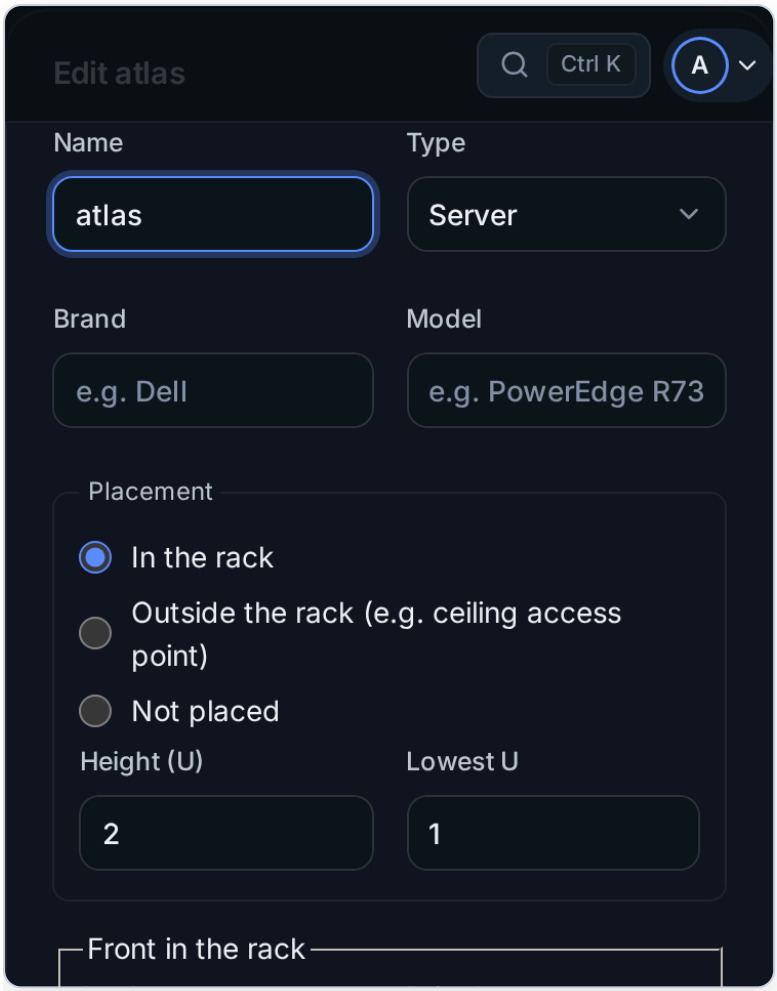
UniFi switches, access points and cameras appear by themselves once the console is added.

Find devices scans your own networks and lists what it recognises (Proxmox, TrueNAS, iDRAC, Synology, UPSes, plugs, Docker and web apps), each with *Add device* or *Add service* to open the form filled in. No password is tried, and nothing is added until you choose it.

Connections

Each device can have one or more ways to read it. Use **read-only** accounts everywhere.

Connection	For	What to create on the device
Redfish	Dell iDRAC, HPE iLO, Lenovo XClarity, Supermicro	A local user with a read-only role
Proxmox	Proxmox VE	A user with role **PVEAuditor** and an API token
TrueNAS	TrueNAS 25.04+	An API key for a *Read-only Administrator*
Synology	Synology DSM	SNMPv3 with SHA and AES
UniFi	UniFi OS consoles	A local *View Only* user without MFA
SNMPv3	Cisco, Aruba, MikroTik, Netgear, QNAP, UPS cards	A read-only SNMPv3 user with SHA and AES
NUT	A USB UPS	NUT's `upsd` listening on the network
Omada	TP-Link Omada	A local *Viewer* on the Omada controller
Shelly, Tasmota	Smart plugs	Nothing (or the plug's web password). Only reads the power.
Home Assistant	Any power sensor Home Assistant reads	A long-lived access token and the sensor's entity ID
Docker	A Docker host's containers	A **socket proxy** that only allows reading containers (`CONTAINERS=1`, `POST=0`)
Portainer	A Portainer environment	An access token, and the environment ID
Ping	Anything without an API	Nothing; it only has to answer ping



Editing atlas: where it sits, how it's drawn, and its connections.

Self-signed certificates

Choose **Pin SHA-256 fingerprint** and press **Fetch certificate**. Lookout checks the fingerprint *before* sending the password. If it changes later (a firmware update), the device stops with *fingerprint mismatch; nothing was sent*. Find out why, then fetch and save the new one.

More

- **Power reading from:** *Automatic* (the controller, then a smart plug, then the estimate), one of them, or *Not counted*. Each device is counted once.
- **Estimated power** for devices without a power sensor.
- **In the rack** with height and lowest U, **outside the rack**, or not placed. **Look** and **Drive bays** change the drawing.

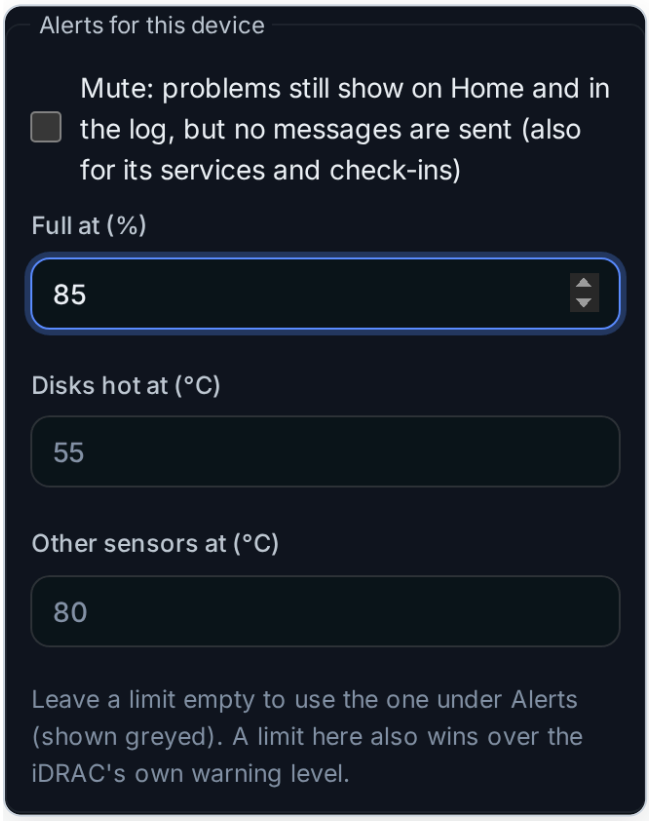
Linux agents

A **Lookout agent** on a VM, container or Linux machine reports what Proxmox can't see: memory as Linux counts it, load, every filesystem, failed and watched **systemd services**, **pending updates** (and security fixes, and a needed reboot) and temperatures. In **Admin → Devices → Linux agents**, **Add agent**, choose what it runs on (a VM from Proxmox, a device or a new one), and run the command it shows on the machine. On the machine: `sudo lookout-agent status`, `test`, `watch add nginx`, `update` and `uninstall`. It only reports, and never updates itself.

Reached through

Each device has **Reached through**: the switch, gateway or host it depends on. *Automatic* uses what UniFi reports (an access point hangs off its switch); for servers, choose it yourself. When that device is down, everything behind it (devices, their services and check-ins) is grouped under it on Home as *Behind core-switch*, their tiles say **Unreachable**, and only one alert is sent, naming what is behind it.

Alerts for this device

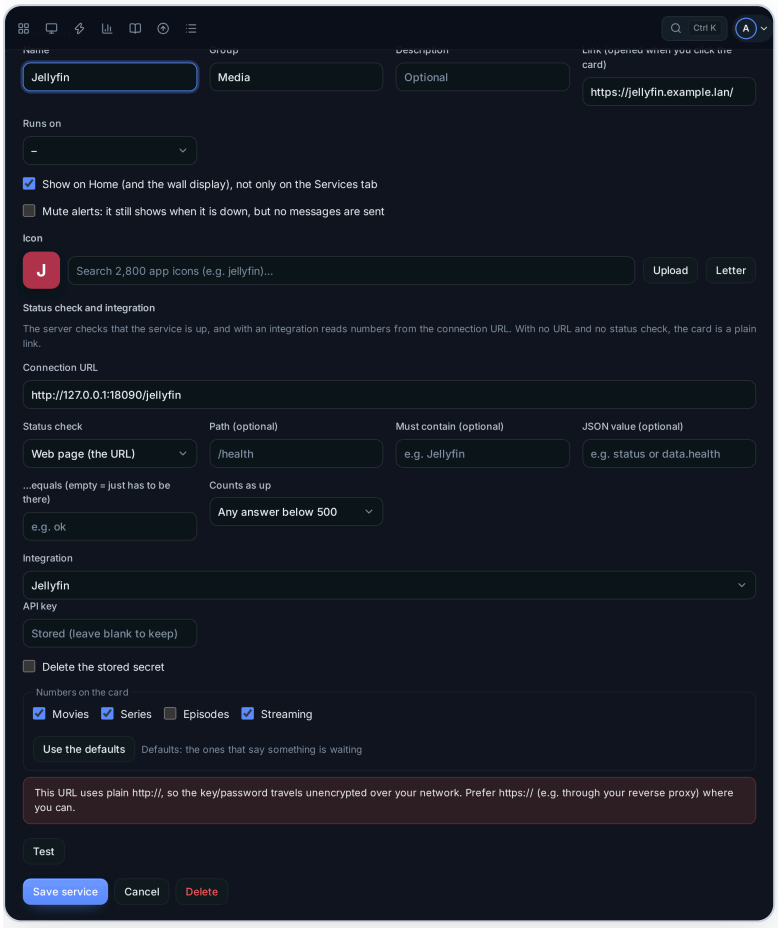


Alerts for vault: its own *Full at* limit; the greyed ones are the general limits.

- **Mute** stops messages for the device and the services and check-ins on it. Its problems still show, marked *Alerts muted*. For planned work, use maintenance instead (chapter 13).
- **Full at**, **Disks hot at** and **Other sensors at**: limits for this device only. Empty uses the limits under Alerts. A limit here also wins over the iDRAC's own warning level.

CHAPTER 11

Admin: services



Editing Jellyfin: name, group, link, icon, connection, status check and integration.

1. **Add service**, then a **name**, **group**, the **link** the card opens and the device it **runs on**.
2. An **icon**: search 2,800 app icons, upload your own, or use a letter.
3. The **Connection URL** (can be an internal address) and a **status check**.
4. The **integration** and its API key, then **Test**.
5. Pick up to four **numbers** for the card and **Save service**.

> Over plain `http://` a key travels unencrypted on your network; the form warns you.

Status checks

Check	Up when
Web page	The URL answers. Optionally a **path** (`/health`), a word it **must contain**, a **JSON value** (`status` = `ok`), and **only success** (2xx/3xx) so a login page doesn't count.
Ping	The host answers ping.
TCP port	Something listens on the port (SSH on 22, a database on 5432).
DNS lookup	Your DNS server (Pi-hole, AdGuard) resolves a name, optionally to an expected address.

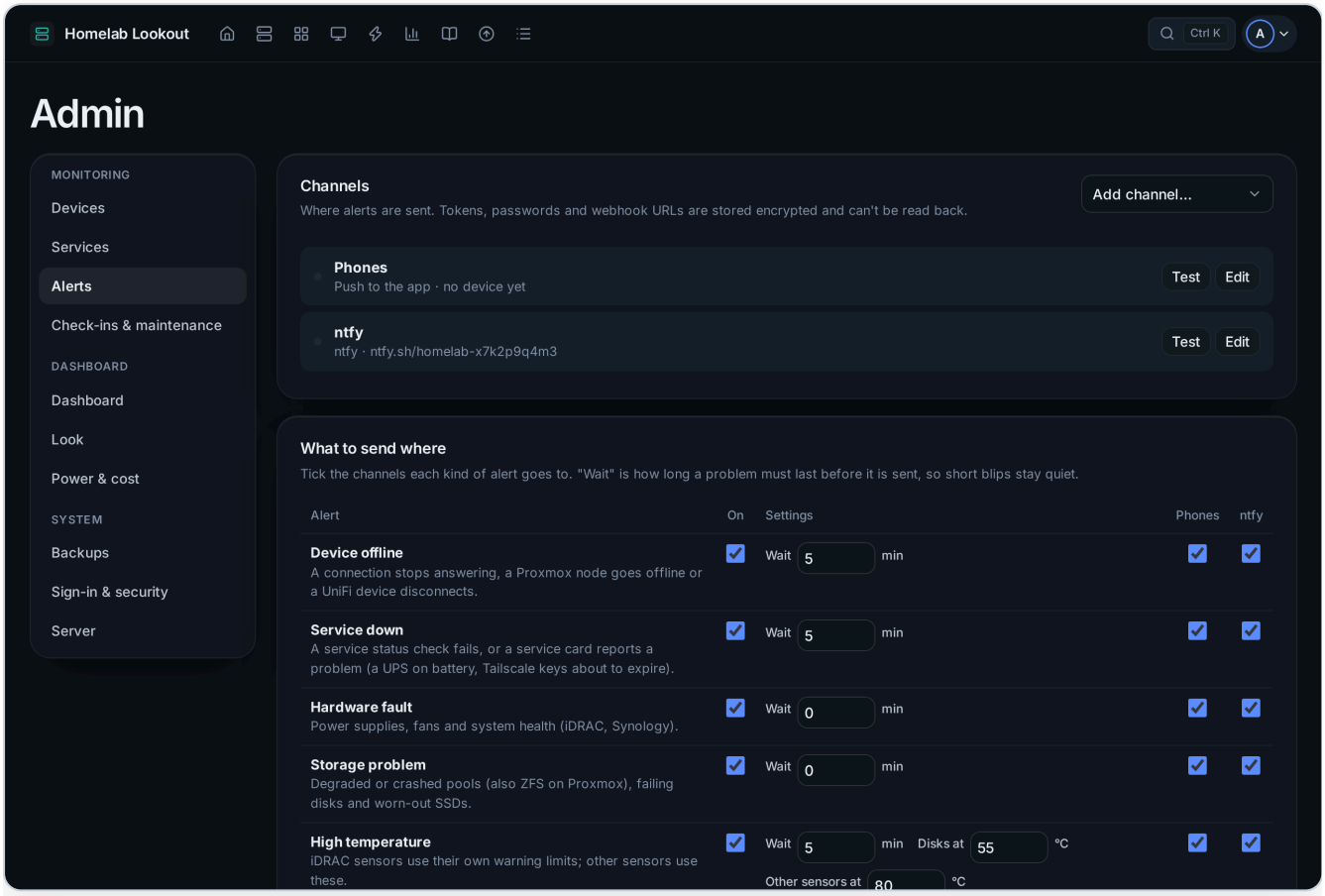
Mute alerts on a service keeps it on the page and on Home, but sends no messages for it.

Infrastructure cards

Integration	Numbers from	Shows
Proxmox VE	A Proxmox device you added	VMs running, CPU, memory; the link opens Proxmox
TrueNAS	A TrueNAS device you added, or its own URL and key	Alerts, unhealthy pools, space
Synology DSM	A Synology device, or DSM's login	Volume used, CPU, memory, temperature
UniFi Network	A UniFi console, or its own login	Clients, WAN, devices offline, speed test
UPS / PeaNUT	A UPS you added, a NUT server, or PeaNUT	Charge, runtime, load, on battery
Tailscale	A read-only API token	Devices online, keys about to expire

CHAPTER 12

Admin: alerts

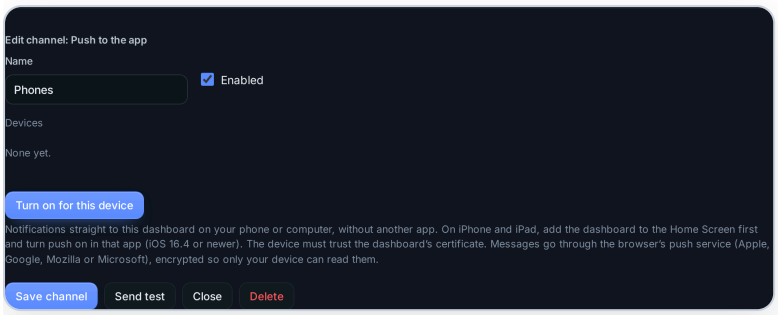


Alerts: channels, and what to send where.

1. Add a **channel** and press **Send test**: push to the app, ntfy, Telegram, Discord, e-mail, iMessage (Sendblue) or a webhook.
2. For each alert type (offline, service down, hardware, storage, temperature, space, TrueNAS, power, container, check-in, backup), pick the channels and a wait time so short blips stay quiet.
3. Choose whether to send a message when it clears, and reminders while it lasts.

Several alerts at once go out as one message. At most 30 messages go to one channel per hour.

Push to the app



A push channel: the devices that get it, and *Turn on for this device*.

1. Add a **Push to the app** channel and save it.
2. On each phone, tablet or computer, open this channel and press **Turn on for this device**; allow notifications.
3. On iPhone and iPad, install the app on the Home Screen first and do it inside the app (iOS 16.4 or newer).

Messages are encrypted so only your device can read them. A critical alert stays on screen, and one open problem has an **Acknowledge** button.

Acknowledge

Acknowledging means someone is on it: the problem keeps showing until it clears, but the reminders stop. Every channel carries an acknowledge link for a single open problem: a button in push and ntfy, a link in Telegram, iMessage, e-mail and Discord, and `ackUrl` in a webhook. The link opens a page that asks first, so a link preview can't acknowledge anything, and it only works for that one problem. On Home, use the clock button on the row.

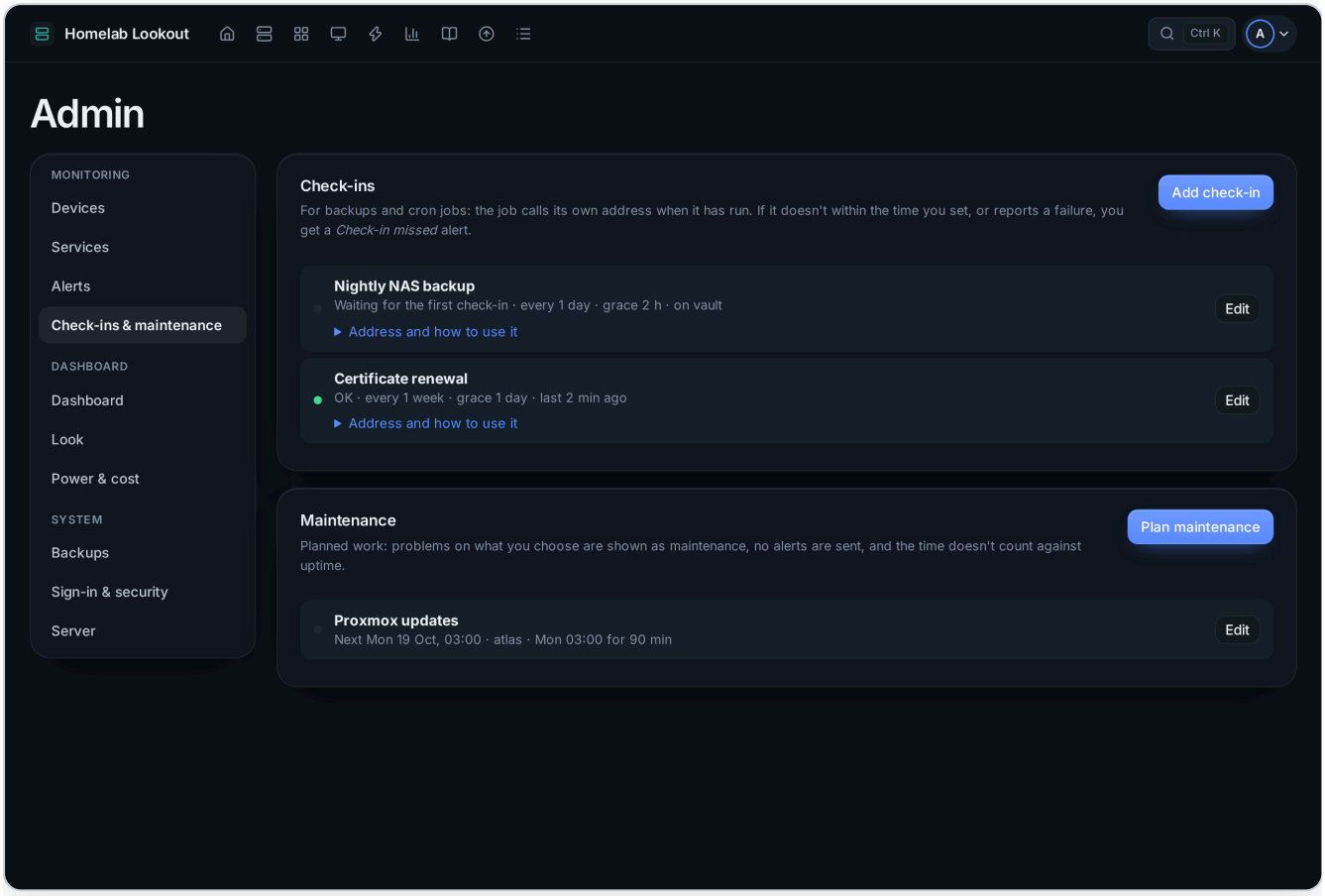
> The links point to the dashboard's address. If you reach it by a name, set **Dashboard address** (chapter 15), so the links work from a phone.

Webhook

A JSON POST for Home Assistant, n8n or Node-RED, with `severity`, `title`, `text`, `link`, `ackUrl` and the list of `alerts`. The URL is stored like a password, because its path is often the secret.

CHAPTER 13

Admin: check-ins and maintenance



Check-ins & maintenance: two jobs that call in, and a weekly window for Proxmox updates.

Check-ins

A check-in catches the backup or cron job that silently stopped running. Give it a name, how often the job runs and how late it may be. Then add its address to the end of the job:

`0 3 * * * backup.sh && curl -fsS -m 10 http://lookout:8080/ping/... || curl -fsS -m 10 http://lookout:8080/ping/.../fail`

It is **OK** while the job calls in time, **Late** after the period and **Missed** after the grace time; `.../fail` reports a failure at once. Keep the address like a password; *New address* replaces it.

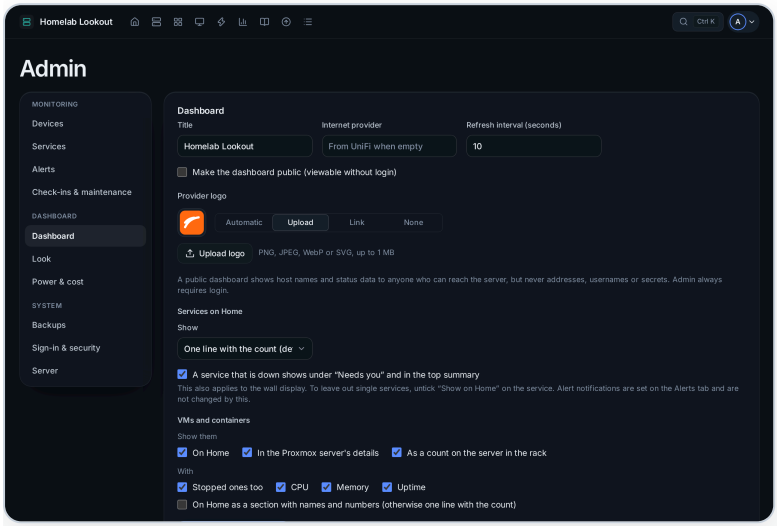
Maintenance

Plan maintenance for work you know about: **Everything**, or the devices, services and check-ins it covers; **Starting now**, **Once**, or **Every week**. While it is on, problems are shown as *Maintenance*, no alerts are sent, and the time doesn't count against uptime. **End now** stops it early.

CHAPTER 14

Admin: dashboard, look and power

Dashboard



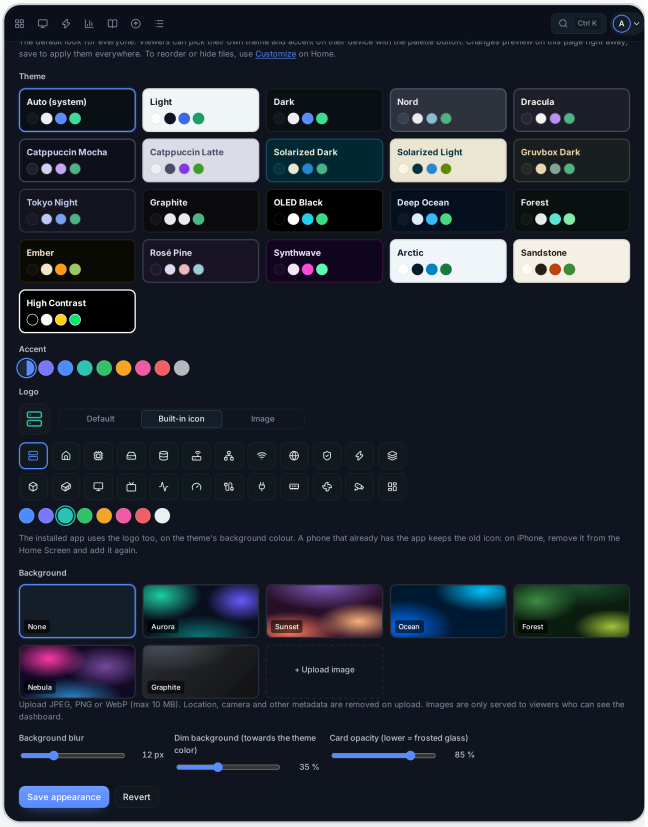
Dashboard: title, provider and what shows on Home.

- **Title**, **internet provider** (and its logo) and **refresh interval**.
- **Services on Home** and **VMs and containers**: where they show and with which numbers.
- **Status page:** switch on the public `/status` page (chapter 5).

> **Keep "Make the dashboard public" off** unless you really mean it: it shows Home, the rack and services to anyone who can reach the server. Share the status page instead.

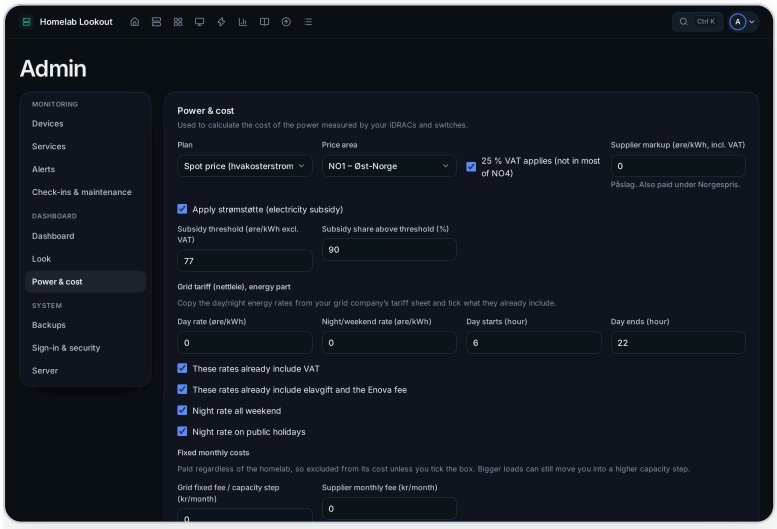
Look

- **Theme** (21) and **accent**: the default look for everyone.
- **Logo:** the default, a built-in icon in a colour, or your own image (PNG, JPEG, WebP or SVG). It is shown in the top bar, on the sign-in page and as the browser tab icon.
- **The app icon** is drawn from the logo on the theme's colour when you press **Save**. A phone that already has the app keeps its old icon: on iPhone, remove the app and add it again.
- **Background** image or gradient, blur, dimming and card opacity.



Look: theme, accent, logo and background, previewed live.

Power & cost



Power & cost: your plan, price area and grid tariff.

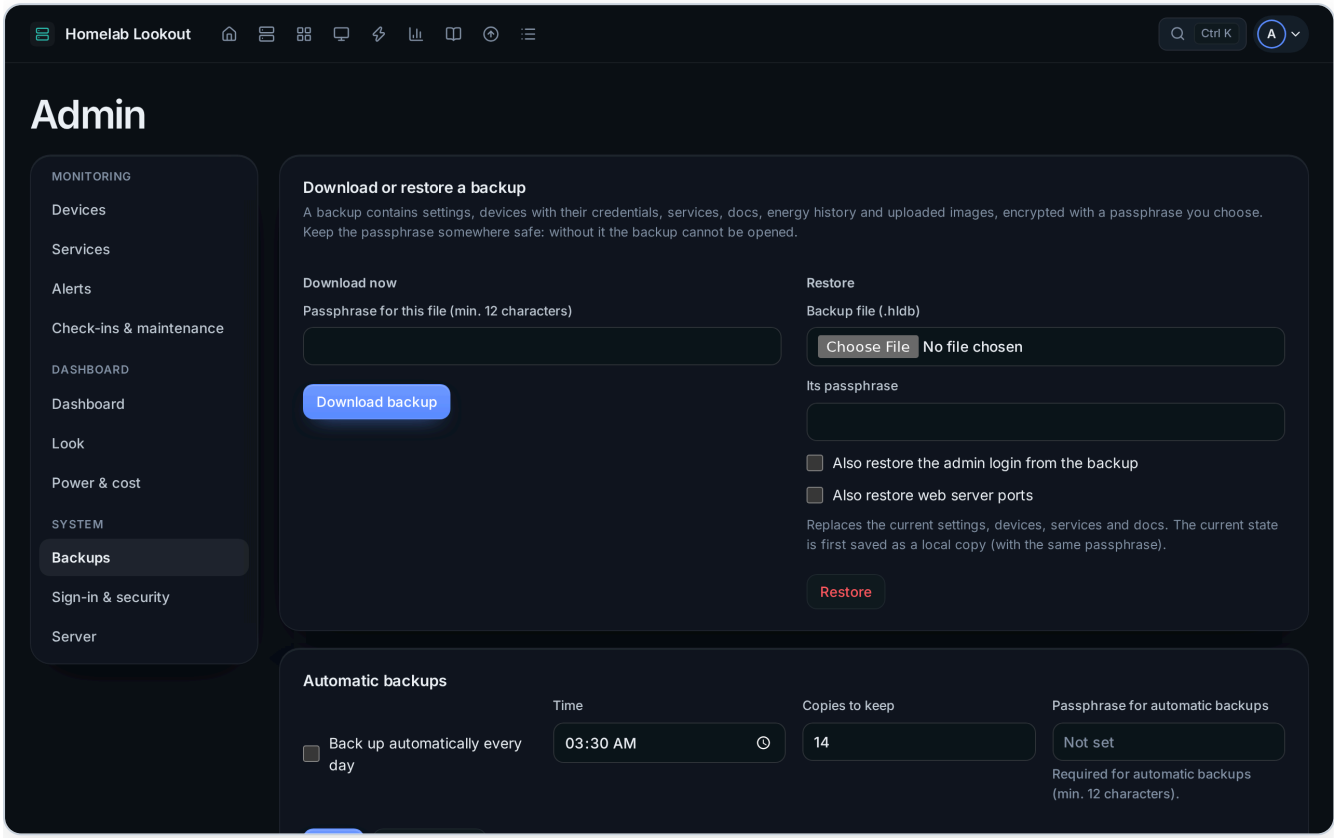
Choose **Norgespris** or **spot price** with your price area, VAT, markup and subsidy. Enter your grid company's day and night rates, the fixed monthly costs and fees.

Include estimated power in cost decides whether devices without a sensor count.

CHAPTER 15

Admin: backups, security and server

Backups

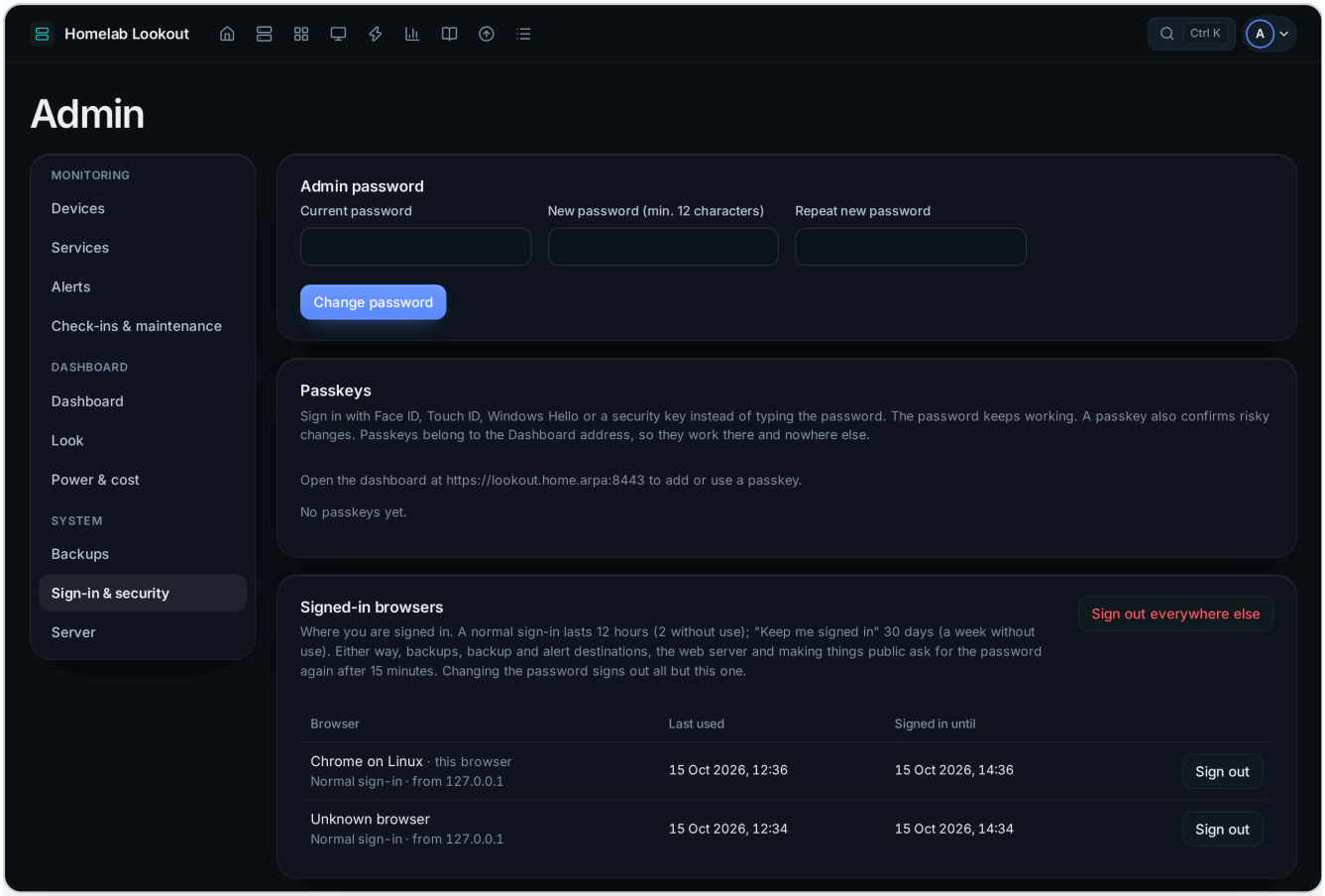


Backups: download or restore, automatic backups and where they go.

- **Download now** gives one encrypted file with everything, including all passwords. Choose a passphrase of at least 12 characters and keep it in your password manager.
- **Restore** reads such a file; the current state is saved first.
- **Automatic backups** run daily and can also go to a NAS folder, S3 storage or another machine over SSH.

> **Moving to a new server:** download a backup, install Lookout on the new machine, and restore it. Devices keep their passwords because they travel inside the encrypted backup.

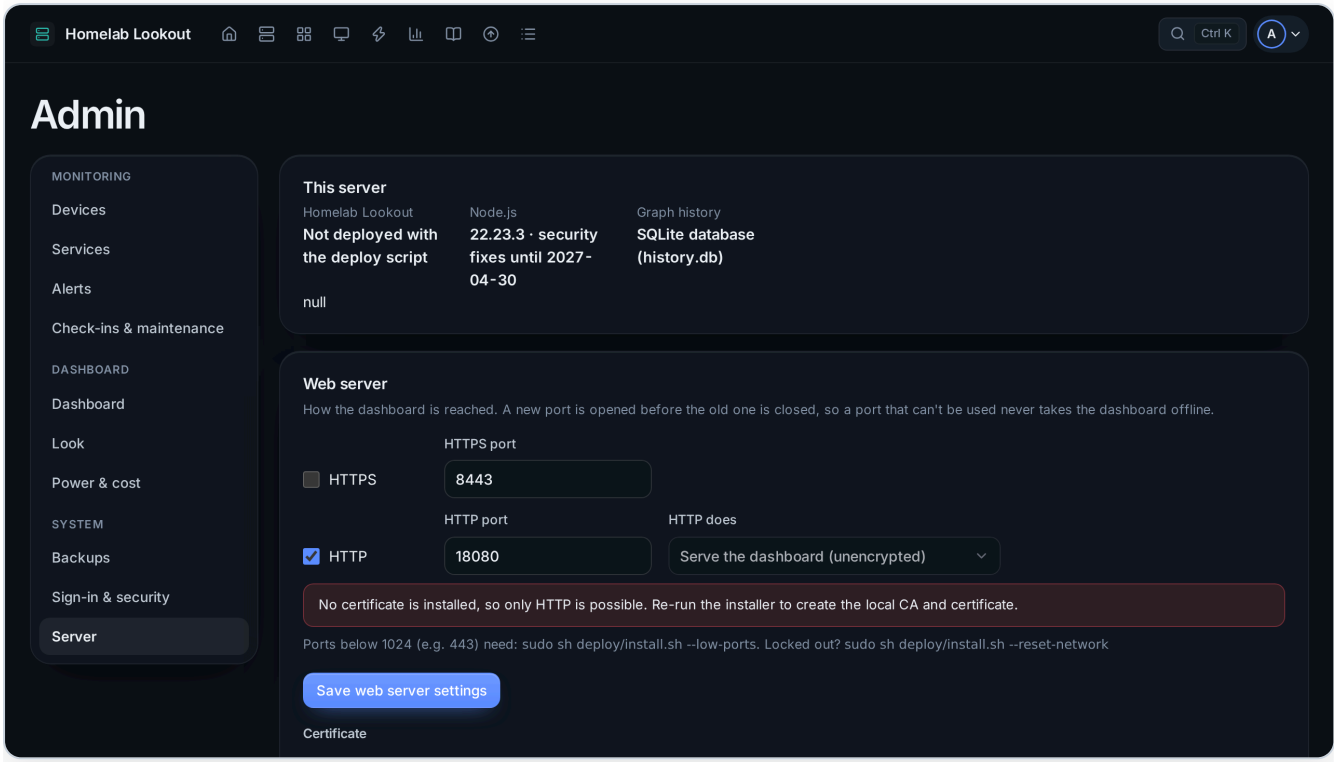
Sign-in & security



Sign-in & security: the password, passkeys, signed-in browsers and remembered devices.

- **Admin password:** changing it signs out every other browser and remembered device.
- **Passkeys:** on the Dashboard address, *Add a passkey*, name it and confirm with Face ID, fingerprint or PIN. A passkey signs in and confirms risky changes; the password keeps working. *Remove* deletes one.
- **Signed-in browsers:** where you are signed in, normal or kept signed in. *Sign out* one, or *Sign out everywhere else*.
- **Remembered devices:** the view-only devices, with *Sign out* for each.

Server



Server: this server's version and Node.js, the web server and its certificate.

Web server: the HTTPS and HTTP ports. Keep HTTP on *Redirect to HTTPS*. **Certificate** shows the names it covers and when it expires, and **Download CA certificate** is here.

Address & reverse proxy

- **Dashboard address:** a domain name, like `https://lookout.home.arpa`. The dashboard answers on it as well as on its IP, and links in alerts, push and passkeys use it.
- **Reverse proxies:** the IP of Nginx Proxy Manager, Caddy, Traefik or another proxy in front of the dashboard. Only these may report the visitor's real address and https, so the lockout and the Log see real addresses.
- **This connection** shows what the server sees of your visit and what doesn't add up. **Setting up a proxy** has copy-ready instructions for each proxy, filled in with your address.

Metrics for Prometheus & Home Assistant

Grafana (through Prometheus) and Home Assistant can read what the dashboard measures: power and energy, temperatures, storage and when it will be full, UPSes, VMs, containers, services, problems and agents. Make a token for each program with the address of the machine it runs on. The token is shown once, with a ready Prometheus config and Home Assistant sensors. It only reads `/metrics` and `/metrics.json`, only from those addresses, and never shows addresses or passwords. The list shows when each token was last used, with **Change** and **Remove**.

CHAPTER 16

Keeping it safe

Lookout was built to be safe by default. A few habits keep it that way.

Do	Why
Keep the dashboard private	Public mode lets anyone on the network see your lab. Share the status page instead.
Use read-only accounts	Then even a leaked secret can't change anything.
Pin fingerprints	The password is only sent to the device you trusted. When one changes, find out why before re-fetching.
HTTPS only	Keep HTTP on *Redirect to HTTPS*.
Keep signed in only on your own devices	If a phone is lost, sign it out under *Signed-in browsers*.
A unique admin password	Change it if someone may have seen it.
Treat check-in and webhook addresses as passwords	Anyone who knows one can use it.
Use a socket proxy for Docker	Never give the dashboard the Docker socket itself.
Store the backup passphrase safely	A backup holds every password Lookout knows.
Look at the Log now and then	Failed sign-ins and unexpected changes show up there.

Secrets are encrypted on disk (AES-256-GCM) and never sent back to the browser. A form only says whether one is set. Changing a device's address or user name without retyping its password deletes the old password, so it is never sent somewhere new.

CHAPTER 17

When something doesn't work

What you see	What to do
A certificate warning	The CA isn't imported on this device, or you used an address the certificate doesn't cover.
No "Install app" option	Trust the certificate first. On an iPhone it has to be Safari.
Can't reach the dashboard	You're not at home or on your VPN. The app opens by itself when it's back.
Too many failed attempts	Wait the time shown. It doubles with each failure, up to an hour.
Asked for the password again	The sign-in ran out, it was signed out, or the password changed. A risky change asks after 15 minutes.
A device says Offline	Admin → Devices → open it → **Test** shows why: a changed certificate, a changed password, or a device that is really down.
A service is red but works	Check its **Connection URL** and **status check** in Admin → Services and press **Test**.
Alerts don't arrive	Admin → Alerts → **Send test** on the channel; check that the alert's row ticks that channel, and that the device isn't muted or in maintenance.
Push doesn't turn on	On iPhone, use the installed app, not Safari. The browser must allow notifications for the site.
Acknowledge links don't open	Set **Dashboard address** to an address your phone can reach.
The app icon is old	Press Save under Look, then remove the app and add it again.
A graph is missing	Graphs appear a few minutes after a device is added.
Excel shows one column	Use the **Excel (NO)** download.

> Every problem, change and sign-in is in the **Log**, and every problem's start and end is under **Incidents**: often the quickest way to find out what happened and when.